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## Introduction to History

We tell the “Story of the Coming of Life” a few days after the Cosmic Creation. This is the second Great Lesson which serves as an introduction to history as well as biology. It is the story of the environment and how it was built up and furnished with plants and animals, with beauty and so many interesting things. It was something that human beings could wonder about and enjoy when they finally arrived on earth.

Our story tells how each life form experimented and how it assisted the next life form. Into the prepared environment came each of the successive agents, the members of the animal kingdom, the members of the plant kingdom. Each species coming with its own particular work to do – so that life was possible for all of the others.

Then something different appeared. It shared some aspect of its functioning with the plants and animals and it also had its own particular gifts. This was the human being. As human beings used their gifts, they have created history. The **history program is integrated into cosmic education. As discussed already, subjects cannot be taught in isolation. They must be inter-linked and interrelated. We have to always think of weaving aspects of biology or geography into a history presentation or sometimes weaving language or math into a history presentation.** This is how the children should be exploring them. The children work together sharing and collaborating according to their interests, which hopefully have been sparked by our stories.

Maria Montessori reminds us that the center of our work is the child. It is not the subject presented nor the teacher. It is the child who needs to be at the center. From an unpublished lecture of Mario Montessori's, he synthesized the following from Maria Montessori lectures on cosmic education: “With others, we believe that in the cosmos, there is harmony, that everything that is in it, both the animate and the inanimate, have collaborated within the creation of our globe, correlating in doing this, their single tasks. But we think that among the innumerable agents who took part in this creation, man has had and has a very important task. Also, creation is not finished and the one agent who as yet has not been taken into consideration is the child. If we take the child from infancy and allow him to develop his psychological power and the potentialities within him, place him in relation with other men and make him realize what mankind has accomplished, then this will form the first step in the formation of the supra-man toward which humanity tends. To the young child, we give guides to the world and the possibility to explore it through his own free activity. To the older child, we must give not the world but the cosmos and a clear vision of how the cosmic energies act in the creation and maintenance of our globe. This must be accompanied by a clear vision of how, through work, the naked and feeble man he was upon his appearance on earth

became the supra-man who has built our present civilization. That is what cosmic education for the older child is. Give him God as he functions in the cosmos to create the earth and give him man, who, starting to live in nature, created the supra-nature in which we live today.”

**In terms of cosmic education, as we look at human beings and the building up of societies and civilizations, we continue the story of the plants and animals, we continue the study of the earth,** of geography – because all of those stories are continuing as well. It's like a tapestry and we have to make all the threads go together, **keeping all of the themes going at the same time.** This is the art of Montessori teaching. It is why we have to keep the stories and these dramatic elements alive in the classroom, why we have to tell these stories every day. If we focus on the story of human beings, language and mathematics will fall into place because they're achievements of human beings – so they will fall into that story of human beings and what they've accomplished and achieved as they have been living here on earth.

As we present the history, we once again need to **take into account the psychological characteristics** of the child and we need to teach according to that psychology. It's only in that way that the work will be interesting to the children. If they are interested, concentration will develop. **If you look around a classroom, whether in practice teaching or observing or in your own classroom, and you see children who are bored, try to identify what aspects of their psychology are not being addressed.** Usually that is the problem. Somewhere or other the psychology is being discounted. If you can identify and correct it, the children will no longer be bored. We should not have bored children in the classroom. There are too many ways to prevent this.

The picture we put before the children is that of human beings as creators – creating through satisfying their needs. They have used their tendencies to do that. Along with their tendencies, they have used their imaginations, so that they were able to make in reality, for themselves, things that had not been here before. They saw how some other animals had long claws for digging and they saw it made things easier to dig from the ground. They realized that we can make something that will help us do the same thing. They used their hands and ability to manipulate to engineer the idea that their imaginations and reasoning developed for them.

What human beings have created has been guided by the intellect – that wonderful gift of human beings. Maria Montessori wanted the children to recognize that many of the human creators are unknown human beings. We need to help the children develop gratitude to the unnamed creators from the past. Who made the first cup or pencil? Who made the first table? Who was the first to compose music or had the idea of spinning thread or of weaving cloth? We certainly know the names of some of the famous inventors and creators of the past – but many are unknown. The inventions remain further developed, built on – but the name of the inventor is unknown.

What Maria Montessori meant by the term supra-nature is everything that has been created or developed by human beings that wasn't here, wasn't provided naturally. Our original environment was worked with and parts of it were used to make all of the things we have now that make our lives generally more comfortable. We've certainly become dependent on this supra-nature. Maria Montessori said it is as if humanity had created a nature above or beyond the nature that was provided by the environment.

Human beings are also the organizers as well as the creators of this nature. Societies evolve for practical reasons. Human beings organized societies. In our story of the coming of life, we have creatures in the story organizing themselves; one cell creatures becoming creatures with more than one cell. Again, in our story, the idea of the cells getting together and cooperating, dividing up labor, developing into organized systems. Just as the first cells organized, so human society has organized itself to be more efficient, more economical. We've helped each other toward perfection, each striving for a self-perfection, showing one another how things can be done better, and as a result, living conditions for many have gotten better and better.

Through that work and organization and the **operating of human tendencies, satisfying those needs has brought about construction of the physical or material plus the spiritual territories** – as described by Maria Montessori.

Therefore we need to show the children, as they study human beings and their place in the cosmos, the life of the past as well as present day life. Because it is only through studying the past that the children can realize what gratitude is owed to those who have come before us. We want to put the children in contact with the intelligence of the common person. If you have studied history, you know it is presented as the study of the exceptional people. Very often, the ordinary people are left out of the story or brushed over lightly. We want the children to realize that it was the common human being who first invented, who first sang, who first made music. The exceptional one, the names we know of, those exceptional people, could not have done their work if the earlier people, the unknown people, had not first done what they did. It is the unknown or unrecognized human beings, with their service and their willingness to serve, who contribute to society.

If we can help the children realize this, we can help them realize that they don't have to be exceptional or a rare person – or conversely that we are all rare – we each just need to be willing to serve to the extent of our potential. If we can help the children realize this, then they will be willing to share themselves with society, with others. If we can help them realize that each person is unique, each one is special and each therefore has something special to offer, they will then be ready to address how each one of us can use our particular gifts to our greatest potential.

September 27, 2004

## History in the Children's House

The foundations for our work in history are laid in the children's house. So let us think of history in the children's house. It is presented according to those psychological characteristics. What is the construction that the first plane child is making? His is introverted and works on individual self-construction. Keeping that in mind, in a general sense, what are the things we need to keep in mind that the child needs to make that construction. Facts – sensorial exploration for example provides facts. Facts need to be facts of the world. The child needs to work individually. The child needs the support and love of family. The child needs to feel secure and self-confident. This enables them to adapt and orientate themselves. They need freedom that allows their own self-construction – freedom to move about and so on. They need to experience order in order to create order within themselves. They need a prepared environment for these things to be useful. They can begin to classify the sensorial qualities of the world. If the child can do this, he will begin to make his own abstractions.

Along with feeling love by family and society, what else does the child need to be able to do vis-a-vis that family? They need to become active members, participating in that family life. In this way they experience the culture of that family.

In this kind of environment that needs to be prepared for the first plane child, we can't help but include history. If we help the child explore his family life, that is history. The story of one's own family is part of that story of human beings on earth, and starts to show us the place of one's own family in that larger society of the world. So there is history all around the child from the moment he is born.

**At the moment of birth, the child begins to participate in that history. The way the family dresses, the food the family eats, the way in which that food is eaten, the stories that he hears from family members, the social graces that are a part of family life – all of these are a part of history.** So the absorbent mind is working there, taking in all of the history of the life of the family that the young child is exposed to. The young child wants to dress the same way, eat the same foods and behave the same way. Clothing, food and language are some of the most basic aspects of family life and human life. They are developments that human beings have been working on since the beginning of their time on earth. These developments have been added to through the ages, embellished and enriched by different people. And all of this becomes a legacy passed on to the child by his family. These become things for the child to explore, first unconsciously and then with the conscious absorbent mind. The child begins to build up his knowledge, his first basic understanding of the story of human beings.

In the children's house, we need to afford the child further explorations about the lives of human beings, not only how they live today but how they have lived in the past. And not only how they are living today here where we live, but how they have lived and live in other places around the world. Whether we speak of today or the past, all of this becomes fact that can simply be presented to the first plane child. We have materials that help us present these facts. There are nomenclature cards about clothing and food, for example. **Initially these are shown and named only since young children don't have a developed sense of time. As they approach the end of the first plane, these cards can also be set out and presented chronologically.** Transportation, for example, might be presented to show the development of time, helping the child to appreciate the sequence. At the end of children's house, the teacher can often times attach dates to these things, after the child has a better idea of time. Books and songs can also be a source of fact for the child in the children's house. Pictures on the walls, stories are useful. Geography folders with pictures of people and animals, food, plants, housing are also a source of facts about humankind past and present. These are pictures we can show them and talk about, so that there is conversation, questions and answers, about these things. Even flags of each country have stories related that can be discussed.

As the children develop **the ability to write in the children's house, they are encouraged to write stories** about these things so their creative writing can be helped to develop by these stories, pictures and conversations.

**None of the presentations are a formal study of history but they provide the factual foundation for the entry into elementary level studies.**

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## History in the Elementary Class

When we come into the elementary class we need to take into account different psychological characteristics. The child is still constructing himself but the means of that construction change. We **no longer have the absorbent mind, we have a reasoning mind**. The child is no longer satisfied with the family and enclosed classroom, he **needs the wider society**, of which his family is a part.

The child needs practice living life in that wider society. To gain security in doing that, the child needs to understand how it came about because in that way it can become known to him and he can begin to understand it. The child is no longer a sensorial explorer. He needs to explore through his reason. And the explorations that he undertakes need to include things that cannot be seen, cannot be sensorially explored. The child now **wants to know the underlying causes and reasons for what has been developed and for what happens in that society and life of the world**.

In order for this kind of exploration to be possible, Maria Montessori said we have to **make use of imagination along with that ability to reason**. We need to be always mindful of the fact that the **child is now interested in the extraordinary**, the magnificent and wondrous. We **need to appeal to that sense of hero-worship**. This is one of the new characteristics of this plane that we need to **make use of in the child's exploration of society, life and history**. We need to also remember the **child needs to explore morality**. Remember that we have talk about how this plane child becomes interested in a sense of right or wrong and a sense of justice. The morality of a group of people, a society, is one of the things that identifies them. The child needs a chance to explore this morality, to understand it, and to come to decide for himself what he thinks is right and wrong, good and evil.

In the **first plane of development, as the child develops the power to choose** his own work, we are **helping him to develop his own will so he is able to choose and accept the rules of society** – to obey the law and live by the guidelines provided by his society.

In the **second plane, the child has to be able to function in the wider society of the world, to recognize himself as an individual amongst a group**. He needs to be helped to work with others after the presentations given by the teacher. He has to learn how to cooperate with others, to share tasks in finding out more information and then to decide as one member of that group what is going to be done with that information that they each provide to the whole project.

The child also needs the opportunity to work by himself when he chooses to. When you look around an elementary classroom, the majority of the work going on should be collaborative or group work. But usually there will be some individual work going on; this should be the minority however of the work. By **encouraging collaborative work, the child is experiencing those factors such as sharing tasks, that human beings recognized were needed to better their ways of life, making them more comfortable and more fulfilled.** The elementary classroom is a practice society. This cooperative society allows the child to touch on the developmental part of history and thus to recognize in himself and through his work with others some of the elements which made possible the construction of the material and spiritual territories. It is through using the group cooperative work that the children can learn most fully from their experiences in the elementary class.

We need to make sure the children understand as we're presenting history to them, that it's their story, the story of their species that we are unfolding to them. It is in this way that we can help them glimpse the part that they can play in this ongoing story. We have the story of the universe and then the furnishings and decorations that prepared the environment for human beings to be able to survive.

This prepared environment was readied for a creature that was different than any other. Human beings have a destiny that it seems is never completely fulfilled in any one lifetime. There is always more to strive for and glimmerings of more that is possible. We can help the child realize there are things for them to strive for. It's this sort of challenge that we want to put before the children in our elementary classes. We show them how the first human beings overcame many challenges, how people throughout time have overcome challenges. And so we can help them see that they too can overcome challenges in their lives.

We help the children see that because they are free to choose so many aspects of their lives they, of necessity, have to be responsible for the choices they make. We can't have one without the other. We can see where total freedom without responsibility leads to problems. Free choice should lead to a discipline in making the right choices.

Cosmic education and history should help the children appreciate these factors and help them decide to do something about them. History is at the center of cosmic education. Because it is our story and because of the nature of our species, history depends on the law and the order of the elements of the created universe. On the plants and animals that conserve that order. History also depends on the recognition of the magnitude of the gifts with which human beings have been endowed - intelligence or reason and love or will. We help the children realize why these gifts seem to have been given in such abundance to human beings and how the universe would have been, in a way, unfulfilled if they hadn't been given. **We are an agent of change with an ability to transform the environment.**

In “To Educate the Human Potential, Maria Montessori said, **“If humanity, which is a fact in nature is going at last to be organized, it will be done only by an education that will give appreciation of all that has been done by human cooperation and readiness to shed prejudices in the interest of common work for the cosmic plan, which may also be called the will of God. This education requires the influence of sacred and deep things to move the human spirit.”** History then is not going to be presented to the children as a chronological series of societies and civilizations. There are two aspects that we will need to keep in mind all year in terms of our history presentations.

The **first is to make sure we are giving history according to the psychological characteristics of the 6-12 year old. The second is to make sure, by the time he is ready to leave us, that he know whatever aspects of history the traditional school curriculum asks for.** So either by 9 or by 12, the child also needs to know whatever aspects are asked for. The child can explore any society, any civilization, any time-period that happens to interest him. The presentations the teacher gives are meant to develop interest and to stir up ideas for further exploration. They also are a way to make sure the information asked for by the public school curriculum are covered. We need to make sure we include those conventional aspects in our stories and presentations. We also need to make sure that if some of the information needs to be memorized, we help **provide the child with material for repetition so that memorization can take place.** At the same time, we need to remember the inter-linking of history with geography, biology, language, math, music, art and so on. We are looking at the changes in the environment brought about by nature and by human beings. Of course all of these will involve all the subject areas.

Maria Montessori said the purpose of history is to help the children develop gratitude to God and gratitude to man. She said about our aim in the introduction to “To Educate the Human Potential”, that we have to **equip our pupils “in their whole being for the adventure of life, accustomed to the free exercise of will and judgement, illuminated by imagination and enthusiasm.”**

Date: September 24, 2004

## The Story of the Coming of Life

Remember the story of how the earth came to be? Today, I am going to tell you the story of the coming of life, of the plants, the animals – and us, human beings.

### Storm

So, go back with your minds to the time when the earth was born as a **tiny little drop of light** and heat, of commotion and confusion. And yet, in this little drop every particle was given a set of laws, so that when the earth cooled everything settled down according to these laws. There was water, the rocks and surrounding them all, the air. The earth looked like a beautiful little pearl. The sun looked at this beautiful little pearl day and night, because it was so beautiful.

### Chaos

One day the sun saw that something was not quite right. Something was beginning to happen. There was trouble. It rained a lot on the earth and, as the water passed through the air, it got mixed up with gas. All of this rain and gas was washing away the rocks, making the sea full of all kinds of salt. There were also a lot of storms breaking up the rocks. The sea was becoming more and more full and the land was being washed away.

### Water

It looked as though once again the order that had been created was beginning to break down. Who was causing this? The sun looked at the water and the water said, “Who me? What have I done? I am made in such a way that **if I become hot, I disappear. If I become cool, I fall down. And being a liquid, I have to go into cracks** – so what can I do? It's that silly sister of mine, air. She stirs me up. It's her fault.”

### Air

Air said: “Me? I like that! I was **given the job of covering the earth with**

**layers and layers of blankets so she wouldn't be cold.** But she has a cold head and cold feet. I have to be always on the run to warm her. And what does the water do? The silly thing, she jumps on my back and takes a ride. It is all right when it is flat, but when there are mountains, I have to climb up. **Water is very heavy and I have to drop her.** I cannot play all the time. I have serious work to do!”

If you ask me, it is all the fault of those rocks. They are so full of wrinkles and bubbles! They have no consideration at all. They don't move an inch to let me pass. On top of that, sometimes they are freezing, so passing next to them I have to cringe; and sometimes they are **so hot I have to climb up to avoid burning.**”

## Rocks

The rocks started to protest: “Blame us, will you? We are not doing anything but sitting around. It is not our fault if we get hot. Do you think we want to get hot?

**When the sun shines upon us, we cannot help it, we do get hot.** What can we do, we are made that way. If you ask me, it is your fault sun – you are the culprit.”

Slowly the earth cooled and its crust formed.

And so it went on. All of them were right, of course. **Everyone was behaving just the way it had been made, according to its nature.** And yet, order was being threatened and something had to be done. What could be done?

## Life

And then, something else was created: a **tiny drop of jelly.** And these had something no other creature had had. They had sensitivity. To exist, they had to eat - each to his own taste, and grow. And they had the power to create others like themselves.

## Fossils

So then life came in the form of these tiny little drops of jelly, each of whom had these laws to follow: eat, grow and create others like yourselves. They were like tiny little machines, and they made other tiny little machines by splitting up. It was as if you had a car and after awhile it split up into two little cars: they grew and after awhile split and so on. **Through these little drops of jelly began the work of cleaning up the sea. As they fed upon the salt of the water, they built their own bodies with them.** Some of these **tiny shells dropped to the bottom of the sea,** and the salts remained trapped in these tiny shells.

## Multi-cellular life and Organs

As time passed and these creatures lived and died, layer upon layer of a new kind of mud piled up on the bottom of the sea like a book, **the book of the earth**. Some of these pages have remained for us to read today, so we can know much of what happened a very long time ago before anyone was born.

These first tiny creatures were made up of just one cell which did all the work: breathing, eating, getting rid of what they could not use to grow, and drifting around in the seas - **cleaning up the water**. Then, as time went on, some of them seem to have said: "Why not join together, because then we can be even more efficient. We can do things better." Along came slightly **bigger creatures, made out of several cells**. They too went on feeding in the water of the seas, growing and creating others like themselves. This went on for a long time until some of them seem to have wondered, instead of each of all their cells doing the same thing, why not give only one job to do to certain cells. It's as if **some cells were told, "you do the eating"**, and others were told, "you do the breathing", and still others were told, "You better be in charge of moving us around." And so then came creatures with arms or legs or mouths or hearts: **creatures with organs**. When we open the book of the earth to the first pages that we can read, all these types of creatures were already there.

*(Begin unrolling the timeline)*

Here we see some of them. Some of these are still alive today. Here we have an **amoeba** with no shell around them - made of just one cell which did all the work. And here we have one-cell flagellates which moved around by using two things that looked like whips. Here are **sponges** made up of many cells, all doing the same job, and the sea anemones, which stayed put on a rock and moved their arms to eat whatever creatures were passing by. One animal of which there were a lot were the **trilobites**. They were like the lobster of their day but they no longer exist today. They were of many sizes – some very big and some very small – but all have disappeared now.

As time went on, all kinds of animals appeared. All kinds of experiments went on. These **cephalopods**, for example, had their **legs attached to their heads**. These **looked like flowers**. But they were **not flowers at all, but stone houses** built by jelly-like creatures. They **took salt out of the sea and built up these stone houses**. These feathery parts were what they waved around in the water so that any little creatures passing by could be grabbed and eaten. Because of their looks and color, they have been called the **sea lilies**. When they died, the tree of rings collapsed and lay at the bottom of

the sea. The water was full of all these different types of life.

Among these there were also some creatures who, instead of eating other creatures, **made their own food. They did this with the salts of the water and sunlight.** They, too, drifted around in the sea and at some point some of them happened upon seeing what it was like outside of the sea. Floating with the tides they got carried to the edge of the sea and there they remained until the tide returned. They found out that the **air outside of the water was full of a gas that they could make use of for their food.** So they **remained there at the edge of the seas**, feeding themselves; and when they died, their bodies remained for preparing the ground for others to come.

## Amphibians

For the first time life was trying out the land. Gradually (show plants on timeline) these are the ones who developed into plants. While these were going on, at about the same time, **two new types of animals appeared in the sea. One was to help the building of the land; those were the corals** – you see them here (point). And **the other was a strange little animal with a rod inside its body.** Until now if the animal had any shell it was on the outside; this one had a rod in the inside. It was the **beginning of animals with bones.** In time this animal developed into the first fish. They were very different from the fish of our day. Once this experiment became a success, a great variety of fish appeared. Some were bug like with great armors for protection. They would remain buried in the mud, and catch other animals as they passed by. Here you see some of them.

What happened? Slowly over a period of time, the **land started to rise and it closed off the seas and oceans.** In some **places with no rain, the land began to dry up.** If you need water for your life and it dries up, what do you do? You must solve the problem. The fish trapped in these places had to figure out something in order to live. Some said: “I will **make a sac inside my body and line it with water.** That way I will **be able to breathe without water.**” They discovered the secret of breathing outside of water. Here came the amphibians, animals that lived partly in water and partly outside of water. The fins became useless so they were changed into legs to walk on land. Animals like **frogs and salamanders** appeared. With them came the sound of the first voice on earth. The silence, except for the sound of falling rocks or rain, was broken. There was now the **voice of the frog.** The amphibians enjoyed their life on the land, for there were also plants. They were growing fast and in great variety, so the amphibians had plenty to eat. There were also a lot of insects around to eat.

## Reptiles

There was just one problem. Because of the way their skins were made, they needed to stay near water; also, for laying eggs, they needed water. Some of them began find new ways to live. They began to be independent of the water. They wanted to go further onto the land. So, what did they do? They **developed a skin that did not dry out**, not even under the sun. **For eggs, they developed a shell**. The new kinds of animals that came out of this experiment were the reptiles. With this new kind of skin that did not dry out, and eggs that did not have to be laid in water, they could move about much more freely and they went much further onto the land.

## Dinosaurs Charts

This was new ground. Nobody was there to stop the, The **reptiles became almost like the lords and masters of the land**. They invaded everywhere - even the air. They grew and grew. As they had bony skeletons to support the weight of their muscles, some of them grew to enormous sizes. Look at this one (*show picture of large dinosaur with little man*). He was about eighty feet long! Can you imagine how big that was? His head was bigger than a human being's. He was so big that he had to have a little brain by the tail as well. His tail was so far away from the brain in his head that the creature could not know if its tail was being bitten - so it had to have a second brain.

Look at this one (T. Rex) ! Can you imagine, when these animals had a fight, how the earth must have trembled? As these huge reptiles moved around the land, other smaller animals had little chance of surviving. **Some of them fled to places these reptiles did not like to go**. These were the **colder regions**; and there they remained. As they were smaller animals, they **needed only a small amount of food** and many of them raided the eggs the reptiles had laid and left. Some of the reptiles did not care about their eggs. They just laid them and went away.

As time went on, these animals developed something to protect their skin and to keep them warm; **they developed fur or feathers and warm blood**. These were the birds and the mammals. What about their eggs? They did want to leave their eggs around like some of the reptiles did so they **carried their eggs around inside their bodies and these were called the mammals**. They birds could not keep them inside, because they had to fly around, so they built nests and sat upon them to keep them warm. When the little ones hatched, they fed and guarded them, sometimes with their lives.

The mammals kept the little ones inside until they were ready to come out; and when they were born, the **mothers fed them with milk**. This was very new. Most of the reptiles did not do this. They left their eggs to hatch on their own. Once they hatched,

they had to look for the food and often were eaten by other animals before long. The parents of mammals and birds stayed with their young until they could take care of themselves. If anyone came to eat the babies, the parents took care of them. This was a new way of caring. Most of the reptiles did not do this.

## Ice Age

By the time these new animals had arrived, the weather had begun to change again. It grew steadily colder and the **reptiles, who had no feathers or fur or warm blood could not stand the cold and began to die** out. The mammals took over. With enough food around, it was their turn to grow big. There were giant pigs and hippopotamuses and elephants with enormous tusks. The weather continued to become colder and colder until **large sheets of ice covered almost the whole earth**. In the end, none of these giant animals survived.

## Human Beings

But then something exciting happened. Towards the end of this very cold period, a completely new type of being appeared. It did not have fur like some of the other animals. Neither did it have huge teeth or huge claws but it did have something which none of the other animals had. It had a **much larger brain with a power to think, imagine and reflect**. It also had an enormous amount of love that was so different that this creature was no longer just an animal. This was a human being. With the human beings, love went beyond their own children. It went to others of their kind, even to those men, women and children they would never see.

It was as though the **earth had to have this long preparation time to make it ready for this special creature**. If human beings had appeared in any of the previous ages, it would not have been possible for them to live. But now everything was ready. If the earth had a voice, she would have said: "I have spread thick carpets of grass, so that your feet can walk on soft ground. I have put flowers in my hair and covered myself with jewels. My cupboards are full with milk and honey, meat and vegetables. Down in the cellars there is coal and iron. Now that all has been made ready, it is your turn to live."

**Date: September 24, 2004**

## **Notes on the Story of the Coming of Life**

This is not a factual story nor a completely made up story. It is a mixture. Interwoven with the facts, there are several ideas. The interplay of life and the environment. There is the idea that life came almost in answer to a problem, as a necessity. Whether or not this is true does not matter. This is one way that Dr. Montessori thought we could make the six year olds realize that life is a continuation of the process that began with the formation of the earth.

Along with the other animals created before it, human beings had a set of laws to follow. This is why the story begins with the air and water and rocks and the sun – the dialogue between them. This emphasizes this idea that everything has its own nature and laws to follow. It is to show there was order and then the order began to be upset and something had to be done.

This accounting of life on earth and of how it may have developed helps the children realize that each of us are part of the human species, that the human species is one part of the earth – and therefore one part of the universe. As Maria Montessori said in “To Educate the Human Potential”, it helps the child to find the universal center of himself with all things, so that his mind does not wander in an aimless quest of knowledge. Everything is part of the same whole entity and that is why it is all-important to see our interconnections. So this is what we constantly need to reflect on.

If we give this framework to the children, then they will know where they fit in. It is natural for human beings at some point or other in their lives to wonder why they are here. What is the purpose of their life here. It is part of our nature to wonder about this. If we can understand the universe and its order, we can understand our place in the universe as well. Then life will have meaning for us. If we can help the children arrive at this point, our Montessori classrooms will really be different than other classrooms. We would have helped the children recognize they are part of a bigger entity and what that part is.

Another important idea in this story is that once life was present on earth, it carried out all kinds of different 'experiments.' It was almost as if there was a guiding unconscious directing all of this development. That is why our story includes little talks,

specialization development, creatures with organs developing. These stories bring out that there was an unconscious intelligence working behind all of this activity. This culminated in the preparation and development of the environment.

Another idea that is important to the story is that by attaching to each life form an instinct for survival, each of them is directed towards what gives it the greatest satisfaction. There is a goal of achievement that goes beyond each creature merely meeting its own needs. While each creature is doing exactly what it needs to do for its own life, it is, at the same time, making things better for others. In one of his lectures in London, in 1958, Mario Montessori said this: “The years we have spent in this work have shown us the enormous change that comes in children, the enormous respect that they develop for creation and every part of it. When they see that stones or colors or animals are simply aspects of the spiritual expression called creation, and the path that life takes is an expression of spirituality because each item of creation is preparing, though unconsciously, a better situation for other creatures. This is what we want them to grasp – not that such and such plant or animal had its origin in such and such a period.”

Within that quote lies our aim for this story. Keeping that in mind, let's consider some of the practicalities. First of all, if the teacher presents this in the classroom, normally we would sit on the floor and unroll the time-line on the floor. So, sit on the floor with the children surrounding. The teacher does not want to overload the story with too many names and terms that the children have never heard before. So one can notice that across the time-line, there are many names, the periods and so on are mentioned in the time-line – but they go unnamed in the first story. In the first telling, the children should get the main ideas, the main milestones in the development of life – not great amounts of details. This includes the progression in terms of efficiency – in function but also in terms of contributing to the environment. This is why some of these primitive creatures are alive today. They are still crucial to the environment. They perform a much needed service.

Finding an effective way of existing by each successive life form resulted also in a progression of independence, in awareness and in love – a spiritual development. Let us think about the main milestones that the teacher wants to be sure to emphasize.

The first is the appearance of unicellular life coming forth as though to prevent the return of chaos on earth – the creatures who came to begin cleaning up the seas. They did this by feeding on the salts that were dissolved in the water – that was starting to become too plentiful. This new element also had the special characteristic of being able to multiply and make others of their own kind. So this is why in the story, that idea is dwelt on a bit with the allegory of the car and the car splitting up. It appeals to

imagination and the second plane characteristics.

The next milestone is the appearance of multi-cellular organisms to do the function more efficiently. The appearance of creatures with organs increased their efficiency. The third milestone is the invasion of the land by the first living things to become independent of the water. Plants, the insects and then the amphibians. In our story it is as though amphibians were forced onto the land because of the land rising up, cutting the water off – so in certain places the land began to dry up.

The fourth milestone is that once on land, there is a continued progression which led to the reptiles, creatures who developed dry skin and shells around their eggs. They were able to live in warm places where there was enough food.

The fifth milestone is the appearance of birds and mammals – creatures better prepared for cold weather because they came with body coverings and warm blood. These new species cared for their young. This was something that many or most of the others did not do. People used to think, historically, reptiles did not care for the young at all; this has changed with recent scientific study.

The last milestone is that out of all of this finally came the appearance of human beings. There we emphasize for the children a bit, how poorly equipped, physically speaking, human beings were. We don't have fur or claws or tusks. But we had something else the others did not have. The ability to think, imagine and reflect – to the extent that we are able – is beyond what any other creature seems able to do. So as we practice the story, these milestones should be kept in mind and they should be emphasized there.

In terms of the factual correctness of the time-line, we note that there are no dates on this timeline. The purpose of the timeline is to provide an impression that all of these things had to happen as the earth was developing. And that it is only at the end of this long preparation that human beings could come and survive. So the teacher's job is to touch the imagination of the children, to make an impression on them. If the children are stirred up, they will begin to look in books or go to an exhibit in a museum and be able to find the facts of current thinking and knowledge for themselves. They will find this approach much more interesting, and what they find will have much more meaning to them than us trying to tell them what is true. Hopefully, they will also come to the awareness that there is not one factual truth to all of this. This is what many people think might have happened. Other people disagree have have other explanations. Ultimately, we want the child to decide for himself what is true and what is not.

September 24, 2004

## Notes on First Presentation - The Coming of Life

This is once again a story for the new children in your class. So you extend the invitation in such a way that you expect them to be there; to the other children in the class, the teacher also wants to extend the invitation but they are free to not attend if so desired. Make sure in telling the story that there is use of mystery, allegory, drama and a sense of wonder. It is in this manner that we can stir the children up. The story is rather long. Children will need to have interest to follow the story throughout. It cannot be merely read in a monotone. **It is in how the story is delivered that the children's attention is held.**

After the story is finished, the teacher wants to **stay with the children a while to let the children examine the timeline** a bit more closely. They will want to see the creatures and ask questions. The teacher should not ask the children questions at this point; this would disturb what the teacher has hopefully stirred up within them. If they ask **something that can be answered briefly and simply, answer. If they ask something more involved and complex, don't try to answer it at that moment.** Ask them to stop by separately to speak about it or to wait for another lesson in the coming days that will address this. The idea is we don't need to satisfy them immediately. There is an expectation in them that we want to nurture – for them to come back for more.

After the presentation, if there is a place on the wall, **hang the timeline up for several days.** In this way, the children can keep coming back to it, looking at it, thinking about it, seeing new things each time. When teacher sees they no longer pay attention to it, take it down and put it away. What happens to something which becomes a permanent fixture? It is no longer noticed. It is better to take it down and bring it out again later to make a stronger impact.

September 24, 2004

## Follow Up Presentations

As one can imagine, this is a material that lends itself to many other presentations. This is an introduction that barely skims over the surface. A day or two later, the **children might want to go over the story again**. This time the teacher can begin to do so piece by piece – **choosing just one section at a time** and providing more facts and more information about it. For instance, we might choose the Cambrian period alone to discuss. We can go into more detail about only that section. When one sees the bands at the top of the timeline, many ideas arise. We might focus on the trilobites; children are intrigued by them. Again, teacher should not be exhaustive but can give more facts and information.

Often the children want to know more about the animals. They love animals at this age (6-7). They are very **attracted to the animals** on the timeline. In a follow-up, the teacher can offer more of the names of the animals. Invertebrates and other new classifications can be introduced. A good book to use along with the timeline is called Life on Earth by David Attenborough. We bring in new names like **cephalopods – creatures with legs on their heads**. Or the gastropods which have their legs attached to their stomachs or to their digestive organs. So we can imagine how this might be interesting with these names. Again, we don't want to be exhaustive but have some carefully chosen books in their environment so we can introduce two or three to the children and then show them where they can find more if interested.

In a follow up presentation, the teacher can pay more attention to these red lines. The red lines on the timeline show the development of certain creatures. If we notice the beginning, we see trilobites. It says “the trilobites developed enormously.” The line describes a trend. Today we have no more trilobites. So if we follow this red line, we see that the line eventually ends. It shows their decrease and extinction. Other things can be observed to start, the corals for example, and continue all the way to the end of the timeline. This represents their presence today. The red lines are not exhaustive or all-inclusive. They are only meant to give ideas and examples.

Whichever aspect(s) the teacher decides to focus on, he should remember to bring in the extraordinary examples in order to appeal to a sense of wonder and awe. Make sure throughout presentations that the idea of each creature meeting its own needs and at the same time contributing to the rest, is present. This is a theme to emphasize. In a follow up presentation, teacher might want to focus on the strips at the top of the timeline – the names of the periods. We might want to provide the etymology of those names. They will see these names again and can understand them now in building engrams up.

We might want to include ages – ages of amphibians, etc. The third row are the periods on the timeline. These names are based on the placement of where the rocks were found. This is something else to offer up to the children.

Often times, the direction the teacher takes is determined by the interest from the children. This is what is hoped for. Sometimes they say nothing. And generally this is a problem to have a group that is too quiet. Initially it seems easy but then realize that no one is questioning, thinking or asking about what teacher has said. Ultimately this is not a good sign. If teacher has a quiet group, it is up to the teacher to bring out the interest and challenges from the children. Otherwise they stop at what was introduced and there is not creative outflow from interest. So we bring out the strange names or whatever it takes to stir them up.

Another area for follow up is climate. Along the top of the timeline is what looks to be ice cycles. They represent ice ages. There were periods of time when much of the earth was covered by sheets of ice. Can we imagine what it was like to live then? How did creatures adapt to survive. What happened in the periods between the glaciers. The ice melted and water levels rose. There is more land exposed during a period of glacial formation. If more water is in the ice, there is more land exposed. In between, the earth also had a chance to warm up and different vegetation rose up.

There is also the possibility that is called the “blank” timeline of life. Even though it is called the blank timeline of life, it is not blank. It has the red lines, and other items from the timeline – but no pictures and no words. It is not used as a matching exercise, where we unroll the two side by side and match them. This chart is meant to be used for memorization. We might pick a period we’ve become interested in and have studied in some depth. We have pieces divided by ages – the pictures and so on are in envelopes that can be manipulated. These envelopes can be categorized in numerous ways but one obvious way is by the period.

Along with the pictures, there are strips with all the names across the top. From memory, the child can build up what the section on the timeline looks like; then the original timeline can be used as a control of error – after the fact of memorization and study. In addition to charts, the teacher is required to make three timelines. The first we have seen in the presentation. The blank one is not required. Usually, teachers do color in the blank one as well, since often the two are colored at the same time to match the same shades and color uses.

(There is no need to turn in the pieces or blank timeline in teacher training.)

**September 27, 2004**

## **The Black Strip**

### **Purpose**

The purpose is to give the children the impression of the length of time it took for human beings to come into existence.

### **Material**

A fabric strip made of black pellon (used to reinforce collars, etc.), heavy duty version. It is seamless, thirty meters long by 15 ½ centimeters wide. At one end there is a red strip 1 centimeter wide. It is a good idea to have a dowel rod about 1 inch in diameter.

### **Prologue**

This presentation comes near the beginning of the year. Most of the children will be 6-7 in general.

### **Begin**

Do you remember when we had the story of the creation of the cosmos? You might remember that we had this great storm of chaos in which everything was very very hot. Most everything began as a gas since it was so hot.

Well here is that point in time (begin unrolling). Here was this big fiery ball of the elements of the universe that had no form as yet – only heat and light and the elements of the universe. Time went on and on as this great storm traveled across our universe. All the particles obeyed the laws of nature just as they do today. Do you remember those laws. One of them was if particles were cold they came together. Another was if particles were very hot they wanted to expand. For a very very long time, this went on. Do you remember that at the center of this vast storm, it was very hot. It took a great amount of time for it to begin to cool. Time went on and on.

Do you remember how hot particles rose to the outer edges of the storm? This happened while at the same time those particles that had risen to the outer edge had

cooled since they came so close to space. And as they became cold, they became heavy again and fell back to the center of the storm. In this way hot particles were carried to the colder parts of the storm and cold particles were carried to the hot parts of the storm. This went on and on for a very long time.

Very slowly, gas particles came together to form liquids. Heavier liquids fell toward the center of the storm while lighter liquids remained at the outer edges of the storm. This went on and on. And yet, still the storm had not begun to look like earth yet.

Do you remember how the crust formed on the surface of the earth? Liquids became cooler as time went on and on. They gradually began to form solids as they cooled and growing heavier, fell toward the center of the storm. This went on for a very long time as well.

Do you also remember how there were hotter substances trapped below the surface that was formed. And there was great pressure built up under the surface that wanted to escape – since elements always follow their basic nature. Hot substances always expand so for a very long time this pressure built up. There were great volcanoes that formed all over the earth. This period took a very long time. It went on and on.

As the volcanoes burst open, substances from the belly of the earth, along with ash and smoke, poured out and filled the outer edges of the storm which had begun to resemble the earth by now. This went on and on for a very long time as well. How long do you think all this took?

It rained and rained as the gases that had been heated up by the substances, arrived at the outer edge and cooled. The substances from the belly of the earth, the ash and smoke that had covered the outer edge of the earth came back down to the earth as liquid. The oceans formed over a very long time. Time went on and on as hot substances became cold and cold substances became hot. Gradually, as time passed, everything found a temperature more in the middle, not very hot and not very cold. As time passed, those substances that fought to escape the belly of the earth found their way to the surface. Chaos that had threatened gradually became more orderly. This took a very long time. But how long do you think it was?

Again, after many many years, the oceans became suitable for something new. Tiny jelly-like animals appeared and somehow managed to find a way to live in the oceans. They began to eat and reproduce more like them and they filled the oceans. This went on and on. Look how much time had passed! (Refer to strip unrolled about  $\frac{1}{2}$  way.)

The tiny jelly-like creatures began to develop in new ways. Over many many years, these one celled animals, like amoebas, began to band together in new ways as they were able to form a new form of life. There appeared multi-cell animals with legs on their heads or on their stomachs. They fought to survive over many many years and gradually they developed new capabilities.

Finally, plants came into being but not before many many more years had passed. How long do you think all this took? Time just went on and on. The plants prepared the way for new animals to try a new way to survive. Some of these animals found their way onto land from the tides and from great masses of earth rising out of the water from underwater volcanoes. There was still cooling going on since not enough time had passed yet for them to all have found their way from under the surface.

Some animals who had developed shells on the outside of their bodies moved onto the earth and gradually developed their bodies so that the hard shell of their bodies grew inside their bodies. Animals began to have bones in their bodies and even the ocean creatures learned how to develop bodies like this that were stronger and better suited to movement – both on the land and in the waters. But this also took a very very long time. Can you imagine how long it must have taken for all this to happen?

The first voice on land was that of the frog. Reptiles appeared after more and more animals learned to leave the oceans and to survive on the land. But this took many many years. These animals developed new skin that did not dry out after leaving the water. We never see anyone develop new skin today since our lives are so short in comparison to the great amount of time it took for the animals to develop and leave the waters of the earth.

Gradually, these animals developed new body parts that allowed them to fly. Great birds appeared over thousands more years like the awesome pterodactyl, a huge flying creature with fingers to grasp its food. We have not even mentioned humans yet since they had not come yet. Time went on and on. But look, now something happened. This small red part at the end shows the very beginning of humankind up to today. Look how much time the earth took to prepare itself before human beings could arrive. Only after so long a time was humankind able to develop and survive. How long did all this take?

It took over 3 billion years for humankind to arrive and our whole history is in this single little red area at the end of this black strip. So we can now see how short our time on earth has been so far.

September 27, 2004

## Notes: The Black Strip

Present this after The Coming of Life and after there has been time for them to reflect on that story. Allow them time to look at the timeline of life and become acquainted with it. Introduce The Black Strip when they are still in the reflective period. The Black Strip brings in a different aspect to these stories. The black signifies the not only the passing of time but most significantly the **unknown**. It is black and without pictures to signify we have no way of knowing for certain what was happening during this time. We can only speculate from the clues left to us by the earth.

The children are left to think “how long did this take”? This piece of material is obviously impressionistic. It is not in exact proportion. When you begin, remind the children of both the story “Creation of the Cosmos” and “The Coming of Life.” Speak just a few points to get them remembering and then ask your key question. Bring out the black strip and begin to unroll it.

Hopefully you will have space to unroll the whole strip such as in a corridor or a well swept sidewalk or the playground. If you don't have such a space or the weather is bad day after day, you can still do the presentation. You then have to fold itself back in itself. Unroll and then fold it back repeatedly.

As you unroll the strip, keep drawing the children's attention to how far it is from the beginning, how much time has passed. That is the impression we want to convey to the children – the passing of time. We are picking up the same points as on the timeline of life but now there are no pictures to distract us from the focus on the amount of time this all took.

You can involve the children as the strip is unrolling, just as was done today. Don't ask for volunteers; call a name. If it's a small group, let everyone have a turn. Present the strip dramatically but don't overact. This lesson is meant to impress the children; it is not a great lesson but a key lesson.

As you go on with the points of the stories and you are nearing the end, begin to prepare the children in advance for that last little bit of strip. Say something like

“I wonder where the human beings are”? Something else is going to happen. The little jelly-like substance should come about halfway through the strip. Figure out where halfway is; get physical markers on the area in which you will present.

Do not just roll up the black strip when you are finished and send the children off. Wait a little, see what their reactions are. Give them time to reflect and to think about your presentation. Do not ask them questions but briefly answer questions they may have.

Make a note to yourself of any interest they have so you can present further stories to them. This piece of material should be brought out again after giving the next great lesson; The Story of the Coming of Human Beings. Then the children will have a chance to be even more impressed: what a vast amount of preparation into the earth before it was ready for human begins and what care and love it took to prepare the earth for the coming of human beings.

So we must experience the gratitude to God, the Creator and all living beings on earth. In this way, we may be moved to give something back.

Maria Montessori first presented this material in India where she developed much of the work in the second plane. There she noted the children often brag about their long history. She reflected and decided that children needed to experience the humility of realizing how short a time human beings had been on the earth.

The first black strip was originally 300 meters long and fifty centimeters wide. It was wound on a long spool supported by two bicycle wheels. In order to make it manageable, she shrunk it down to one tenth its original size.

Show the children how to roll it up. Start at the red strip, watch them so it is nice and neat. We are expected to make this material. Make sure the dimensions are saved in case you have to remake it at another time in the future.

September 28, 2004

# **The Story of the Coming of Human Beings**

## **Prologue**

A little time goes by since The Coming of Life and the Timeline of Life. The Black Strip would have been presented. Give children time to reflect after each. This is another Great Lesson. It is an introduction opening up another area of study for the children. Gather up children new to the class (mandatory) and invite others as well.

## **Begin**

Do you remember on the timeline of life, right at the end of that timeline there was a picture, a picture of a human being. When we had that timeline, we talked about the earth settling down and the plants and animals coming. Do you remember this on the timeline?

## **Long Preparation**

Life began to appear on the land and eventually big animals came. There were frogs and something special about them. It was their voice. And birds began to fly about. Most of them built nests to lay their eggs in. Then, the earth was finally ready. At the very end of that timeline, we find human beings. A human being just like you and me, first appeared on earth.

Then do you remember the other day we had the Black Strip? We saw how long long long it took before human beings came? Do you remember that long time? Just that little red strip at the end is the time that human beings have been on earth.

Today, I'm going to tell you a little bit more about the story of human beings and what has made them so special.

## **Thought and Reflection**

When human beings came, they were very different from the other animals

that came before them. Do you remember? We said that before, because that is important. What made human beings so different wasn't just the shape of their body, but their minds. Their minds could think just like our minds can think. You're probably thinking right now, while I tell you the story. Maybe you are wondering what I'm going to say next. Maybe you're already making up the next part of the story for yourself in your own mind.

That's thinking. A mind that can think about all kinds of different things, that can plan for the future and that can reflect on the past, is one of the differences between human beings and the other animals. Human beings could see the sun and could feel the warmth of the sun. They could feel the wind blow and the rain come down. They could wonder about these things, wonder why they were happening. What made these things happen? That means that human beings could think about them.

## **Story-Telling**

At night, they could see the stars up in the sky and the moon shining in the sky. They could wonder about these things also. Then they also did something else. They began to make up stories about these things. So the sun had a story and the moon had a story. And some of the stars had stories. The people who made up these stories told them to the other people around them.

## **Love**

The other thing that made human beings so different was that they could love. We can love also. I can love my sisters and my brothers, my husband and my children, and my mother and my friends. If you have sisters and brothers or a mom and dad or friends, you can love them too. I bet you do, don't you? But we can also do something else. We can wish that all of the people all over the world have enough food to eat. We can wish they have a safe place to live. This means that we can love people that we don't even know, that we'll never meet.

## **Will**

With these two great gifts, a mind to think with and the ability to love, human beings, when they came to live on the earth, were able to do a great many things – more than the plants and the animals could do. Each of the other animals was given a particular life that belonged just to it. Each plant was given a particular kind of life that belonged just to it. Cows eat grass – they never eat meat, only grass. Squirrels eat nuts. This is the kind of food their bodies can handle. Some birds eat worms and others eat seeds and different kinds of birds eat different things. But human beings can eat all kinds

of different things.

## **Food**

For breakfast this morning, I had cereal with strawberries and milk and a cup of coffee. All kinds of different things. Do you remember what you ate this morning? Was it different kinds of things? Yeah.

## **Shelter**

Most birds build nests for their homes. Some human beings live in tents. Some live in apartment buildings. Some live in houses. Some human beings live in houses made of bricks. Some houses are made of stone or wood. Some of them have wooden doors and others have steel doors. Some even have glass doors. Just think of all the different kinds of houses that you see just on your way to school every day.

## **Hands and Standing Upright**

Human beings were given something else also, which was another special gift. Human beings were given hands which can be used to make things. When human beings came onto the earth, they were able to walk on two legs, freeing the hands to do things and make things. Just imagine us here at school. If you had to walk around on your hands and feet, how would you carry your work to your table. It would be difficult, right? How would you water the plants or hold your pencil to write?

## **Each Human Builds the World**

We have to remember that human beings are very special because we have those special gifts of a mind to think with and a special way of loving, and our hands. And so we are special, each one of us here, sitting and listening to this lesson. All of the other children in the class are special also. And all of the children in the school are special. And all children, and all men and all women, all over the world, are also special. That is why we have to be so careful about how we treat each other – because we are each special. There is a specialness about each individual person.

## **Ancient Humans Built Our World**

The story of human beings, the story of men and women and children began

a long long time ago. And it is a very exciting story with a whole lot of things to find out about human beings and what they've been doing since they arrived on the earth. Another day, we'll have another story to find out more about these human beings. Just remember that those earliest human beings were just like us. They just lived a long time before any of us were born.

September 28, 2004

## The Hand Timeline

### Prologue

A few more days pass since the last story. The children have time to reflect. You would call a few children together for the next presentation. The **Coming of the Human Being is the Great Lesson. This is one of the key lessons that would be gone into that picks up on that story.**

### Begin

Do you remember the other day when we had the timeline of life? Remember that at the end, after the plants and the animals came to grow on the earth, and to begin to move around on the earth, finally after a long long time, human beings came. There was finally a place ready in which they could live and survive.

And then a few days after that story, we had the story about the coming of human beings and about some of the things that make them special. They were able to find their own food to eat. They were able to make their own shelter out of stones that they found lying around. The sun and the water and the earth made it possible for them to live.

Today we're going to have another timeline to show us something else about human beings. This timeline is about men and women and children, about human beings just like us, and about all of the time that human beings have been living on the earth. So today we're going to look at this timeline. (gradually begin unrolling)

Remember we said that it is a timeline about human beings. Human beings were living during all of this time. And because they were human beings, we know they must have been thinking and planning and reflecting; those things that our minds can do. But we don't know exactly for sure what they were doing during all of this time, living their lives, finding food to eat, making shelter for themselves. Time went on and time went on, and human beings were living here during all of this time.

Now are you watching? Here we have a special picture. It's a picture of a

hand. A hand holding a stone tool. Remember how important the gift of the hand is? We don't know for sure what human beings were doing, but we know they must have been using their hands to catch fish in order to eat those fish, or to dig food out of the ground.

They must have used their hands to make some clothing for themselves, because remember they came when it was cold on the earth. Time went on and on. And human beings were here during all of this time, planning for what they would do the next day, thinking about that storm that passed over last night – using their minds and hands – satisfying their needs so they could go on living. But during this time, we don't know exactly what they were doing or how they did it.

(unroll until here red strip appears at the end)

It's only when we get to here (red strip appears) that we can know for sure what human beings were doing. Because it is only from this point in time that we have written records. During all of this other time, we know human beings had to find food to eat, or their bodies wouldn't have survived. We know they had to find or make shelters to protect themselves from the elements. We know they made decorations for their bodies because we've found some of them.

But what we don't have from all of this time is any books or magazines or newspapers telling us what they did. All of this time that human beings were on earth, we know they must have been using their special gifts, the mind that can think of the special way of loving and of their hands. We know they had to find food. We know they had to meet those needs all human beings have for clothing, for shelter. The only records we have from all of this time are some of the artifacts, some of the objects that have been found that they made, or some of the remains of their houses. There is no book written by any of these people telling us exactly what their life was like. It's only when we get to that little bit of red at the end that we can know for sure what some of the human beings were doing. It's only from that amount of time that we have written records, writing left for us by earlier people telling us some facts about what their lives were like.

We're going to have many more stories about this period of time during which there has been writing. But those will be stories for other days.

## **Epilogue**

The story of human beings includes pre-history and history. They are both aspects of the story of human beings we will go into with the children. During both periods of time, humanity was using their hands. This timeline emphasizes that point. This timeline should also strikingly show the two periods of time, between the period of unrecorded history and recorded history.

September 28, 2004

## Notes for the Hand Timeline

This is a story that is told to the children to **aid their imaginations**. So the material itself is very impressionistic. It is giving an impression of this immense amount of time, the time of unrecorded history and recorded history. It's also considered a Key Lesson because it brings out to the children two points that they need to know about - the fact that the hand is the instrument of the mind and the fact that the amount of recorded history is very very short compared to unrecorded history.

We also bring in the idea that we only know real facts when we get to the time of recorded history. Before that, what we think and have pieced together – we don't know those real facts. So it's important that as the children begin studying history that they realize there are these two aspects to it. They need to think about both periods of time.

Notice there aren't any pictures on the timeline except for the hand – just a hand holding a stone tool or a stone implement. A little later in their time in the elementary, the children might want to pick up on the stone tools and look into the stone cultures. What was unique about each of them? One thing they will learn is that different stones were used for different things. Some stones are better for one purpose and others are better for another purpose.

The Great Lesson, “The Coming of Human Beings”, is the introduction of the story of human beings. Out of that story, we'll pick up on various key points with little stories that become Key Lessons in the classroom.

**A Key Lesson means it includes a new point of awareness or a new help to understanding that the children need in building up their knowledge and understanding** about how things work.

**A presentation like this also gives the imagination the possibility of reconstructing for itself what each individual might think happened.** Each of you might visualize it a little differently, with your own minds and imaginations. If we provided all kinds of pictures on this timeline, the work of the imagination would already

be done.

Maybe now you'd like to draw your own pictures, showing how you think some of these early human beings got the food they needed to eat, or how you think they made shelters for themselves. What each of you would draw would probably be different since we aren't beginning with pictures, but only a story that sparks the imagination. We want the children also to be sparked in imagination.

**Our hands are really the first tools that human beings most probably used since they are with us all the time naturally. Don't we still use our hands to make a cup, perhaps to hold water to splash on our face?** We can imagine the first human beings using their hands as cups to hold water as well. They didn't have glasses to drink from and probably used their hands for this as well.

There is a good book called "The Hand" by Frank O. Wilson. He traces how human beings have used their hands, how the hand and mind work together to aid the development that humans have made.

It's **important that we make the children aware of this long time of unrecorded history** because today, information is so readily available. We have books, televisions, the internet; we can not only read about what happened yesterday, we can find out what is happening right now on the other side of the world. The children should realize it has not always been like this.

This presentation is **one of the ways in which we can also get the children interested in writing**. It is only when we get to that red strip at the end that we have writing. Some of the children are still struggling to develop their handwriting and through a lesson like this, we can help to develop their interest in writing. We wonder who the first person was who wrote the first letter. That is someone we don't know but are grateful to. This ties in with the idea of gratitude for unknown human beings.

If we change the story, we need the special gifts and Fundamental Needs of Human Beings. The Coming of Human Beings can also be changed: special gifts as well need to be kept.

October 13, 2004

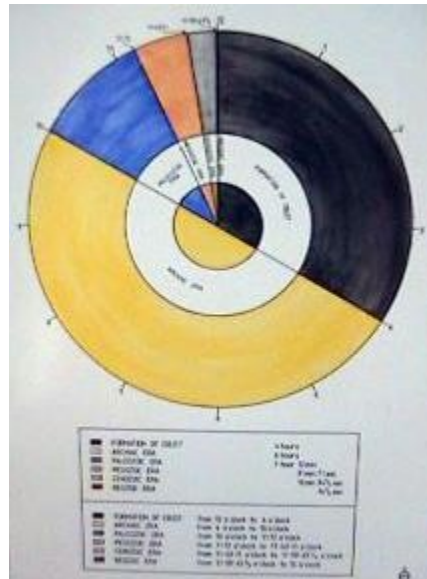
## Clock of Eras

### Prerequisites

The children need to know how to tell hours and minutes on the analog clock before doing this work.

### Material

A chart representing a 12-hour analog clock with colored areas symbolizing the periods of pre-history.



### Material Making

Colors beginning at 12 o'clock - are black, yellow, orange and gray. These match the Timeline of Life. The **oranges need to match on the Timeline and Clock of Eras**. Therefore, do the Clock of Eras and the Timeline of Life charts at the same time.

### Prologue

This story **comes after the Black Strip and is more technical**. We are trying to intrigue the child and get his imagination working. This is another way of

presenting the Timeline of Life and the Black Strip.

**How long did that preparation of the earth take, before human beings arrived?**

Until now, we have not formally talked about time nor referred much to the geological eras. We gather a small group 6-8 children.

### **Begin**

Remember, a while ago, we had the Black Strip? We talked about time going on and on. Do you remember?

We said what a long time the earth had been here and what a short time human beings had been here. We talked of what preparation had been needed so that human beings could survive on earth.

This time, instead of a long black strip, I have something that looks like a clock. (hold up new chart – Clock of the Eras)

But look at it. It's got numbers like a clock – and also colors. We'll find out about the colors and what they mean. What if we took all the time, from the beginning of the earth until today, and shrunk it down to fit on a clock? Do you think you could imagine that? All of time fitting into 12 hours on our clock here.

Starting here (point to 12), going all the way around, that's all the time from the beginning of the earth to now. So you can see the clock has numbers and here we start with 12 as the beginning of the earth.

Remember we said in the beginning it was very very hot. And it took a long time for everything to settle down and cool off?

It took all this time (point to black area of clock's circle) for the first development. A **crust formed and the earth cooled** down. And so the water settled down too. It no longer jumped off the earth into gas. So, this black area is the **formation of the earth and its crust**. It took from **12 o'clock to 4 o'clock**.

And then, all the **waters on the earth collected**, sometime around here (point to yellow area), and little jelly-like creatures appeared in the waters. The one-celled creatures came. Remember those – when the amoebas and the trilobites lived? All of this time – from **4 o'clock to 10 o'clock** is called the **Archaic Era**. This is the time **when tiny creatures came into being**.

From **10 o'clock to a bit past 11 o'clock** was the time – also very very long ago, when **animals were living in the seas**. Remember the cephalopods? This is the time from 10 o'clock to a bit past 11 o'clock that we call the **Paleozoic Era**. 'Paleo' means “very very long ago” and 'zo' means life. So this is life from very very long ago. All of this life at this time lived in the water until the very end of this period of time. But we're not finished yet.

We have this time from **a bit after 11 o'clock to just before 12 o'clock**. This is the time when those **reptiles came onto the land**. You all remember those dinosaurs. This era is called the **Mesozoic Era**. 'Meso' means **middle**. So this term means middle life. Toward the **end of the Mesozoic era is when the mammals came onto the earth**.

Here, **just before 12 o'clock to not quite to the end of our clock** – is the **Cenozoic Era**. 'Ceno' means **recent**. So this term means recent life. This is the time of the **huge mammals like great elephants and great pigs**.

Now you have to look carefully. Here, at 11:59 and 45 ½ seconds, that is when the **Neozoic Era** began. 'Neo' means **new**. So this term means new life. Can you see this thin red line. That is when a new life began.

Do you know what that new life is? It's us – human beings. **Human beings are the new life of the Neozoic Era**. These last few seconds, compared with all of the prior time – that is all the time human beings have been on earth. **That's just 14 ½ seconds of a 12 hour period**. How many seconds are there in a 12 hour period?

So again, it took all of this time for the earth to become prepared for the arrival of human beings. I'll leave this chart out for a while. It stays in the chart box under history.

## Epilogue

Leave the chart out for a while. The children will discuss it and give their opinions and debate amongst themselves regarding the information on it; this is a language connection. They can see how this ties to the Timeline of Life. It may even be an incentive to revisit the Timeline of Life again. We also refer to the Black Strip; they may compare the Clock of Eras with the Timeline of Life or with the Black Strip. They may even get all three out to compare all.

This time we **articulate the names of the eras**. And as with so many new terms, we also provide the **etymology**. We know etymology helps them to remember the names.

There is a legend at the bottom of the chart. Don't mention it unless they ask. The top legend tells the name of the era, followed by the duration in hours and minutes (based on a 12-hour total duration). The bottom legend tells the name of the era and the time on the clock, for example, the Archaic Period runs from 4 to 10 o'clock.

Another fact that can be revisited along the way of research and study of civilizations, is that as we approach more recent eras, **developments seem to speed up**. Things seem to go faster as we approach the present. We see this in human history. It took a long time for writing to be invented and also a long time for civilizations to rise – and then for cities, huge urban centers to rise. It seemed not so long ago, people were afraid of sailing off the edge of the earth. And now, we travel into space and back.

Why is this case that things seem to have sped up?

On the Clock of Eras, in the last 15 seconds on the 12-hour clock – not even a half of a minute – is when all of human history has existed. In presenting this, one should keep in mind the idea that human beings could not have come and survived before they did. We can feel a sense of gratitude for all the development that took place before we arrived. It's the same with prior civilizations. If not for the people who came before us, we would not have the modern civilizations we have today.

October 13, 2004

## First Timeline of Human Beings

### Material

Timeline of human beings. This is to be unrolled in a lesson framework.

### Prologue

Present this on the floor with the children next to you.

### Begin

So, we had last talked about the point in history when human beings came. Remember we saw the **end of the black strip and the little red part at the end? This was the part when humans first arrived on the earth.**

And do you remember the Black Strip? Only at the very end of it did we see that human beings had arrived on earth. It was a tiny part at the very very end.

### Mind and Will

Do you remember we said that human beings had **two special gifts** that made them different from other creatures. These are our **minds** used for reasoning (which were greatly developed by the use of our hands) and secondly, the ability to **will**, to love.

### Hand

We spoke of how the hands were used by human beings to do things. Remember the little red strip on the Hand Timeline? Remember, it wasn't until *then*, when we saw the red strip, that people could write things down. Before that, we could not know for sure what they were doing to communicate in those days. But we can guess about these things from the clues they left.

They had to have houses for shelter and they must have needed food. But we don't know for sure how these things were done. **We can only look at the clues and the evidence and guess about these things.**

Today, I have another timeline. It is a **timeline of early human beings**. It is about the time just before writing. **(begin unrolling the timeline without speaking)**

### **Ice Age**

What do you see? What do you notice? Yes, human beings. And yes, there are huge chunks of ice. When humans arrived, there was an ice age. That means that ice covered most of the earth at this time.

### **Clothing**

What else do we notice? Yes, the people had no clothes on. These people could not just go to the store and buy clothes. They had to make up a solution for clothes. So they appear to have used the furs of animals. This means they had to catch the animal, too.

(continue unrolling)

### **Shelter**

Something else we notice is that there are no houses. These people used caves to live in. They didn't have to build their own houses in that case. They just used what other animals also used to live in. Human beings could observe and copy other animals.

### **Food**

What did they eat? They couldn't go to the grocery store like us. It looks like they were eating meat here at this time. See this man? He's picking up a rock. He's probably thinking about how to get his food. Maybe he's wondering if this rock would be good for making a sharp edge to cut with. Maybe he's afraid the rock will only crumble when he uses it to catch his food or to fight off an animal.

### **Tools and Fire**

Down here, we see someone scraping the skin of an animal. Again, they are using a tool to do this. And down here, we can see that someone figured out how to make fire – and to control fire. Think of that! Can you **imagine what it would be like to discover how to control fire for the first time!** We might have had to go into the cave with no light. But with this fire, we could now carry it on a stick to find our way and to avoid other animals in the dark cave. Not only that, humans discovered they could **cook with fire**. Before that, we ate food without cooking it at all.

(approaching the end of the roll)

### **Community and Cooperation**

So we see people on this timeline, looking at their hands and thinking. We see them cooking. We see pictures of people working together. Here's some people cutting the fur off of an animal they had captured. These people figured out that if they worked together they could do more and do greater things. Maybe today I'd catch a rabbit for myself. But then tomorrow I'd have to catch another all over again. But perhaps if I worked with others, I could catch something larger and eat for several days.

So, I'll leave this timeline out. Whoever the last person to look at it is today, please come to get me and we can put it up properly. I want you to know we begin rolling it up at this end (point to the end). So that's my story for today. On another day, we can talk about this stuff some more.

### **Epilogue**

A first presentation has special qualities. This is another timeline to leave out a while before rolling it up and putting it away. The children can look at and ponder it for a time.

There should be some books on the shelf about early human beings for them to explore further and to generate questions. They can ask for more stories about a particular aspect of the timeline.

The first human beings lived differently than we do. They lived so long, long, long ago but in many ways, they live the same as we do today. They fed themselves. They clothed themselves. They needed shelter as we do. And they learned to cooperate with each other so life could be easier for the group of people to which they belonged.

Parts of these themes were hinted at here in this lesson. But we should also start a dialogue about others themes. This tells one what the children are interested in. Also, be sure to **cover the Fundamental Needs of Human Beings**. Here, we are beginning to take up the building of the human sphere on earth.

There is writing across the top of the timeline. Don't bring this out on the first lesson but merely respond if asked. **Bring out the writing and other factors at subsequent lessons on this timeline.** Why do we have a lower and upper Paleolithic? It has to do with digging for artifacts. The first layer of artifacts are used to denote the upper Paleolithic period, whereas the lower layers of artifacts would comprise the lower Paleolithic period.

Icicles represent ice ages. Dark dark blue is not water, it represents cold ice. Then warmer time is represented by the red line when the land was exposed. Where did the ice extend to? So, red represents exposed land.

Vegetation is another thing to bring up. We have Tundra, with its cold sparse vegetation. Steppe lands with grass supported large herds of animals such as reindeer. This is another thing to raise with the children. Maybe they would be interested in knowing what animals were present in those times. How were those animals related to human beings in that time?

Another thing to examine is the stone tools. They begin without any great detail in the chart. We do the same – simply stating that people had tools and providing some idea how they were made and used. But for 9 to 12 year olds, they might go deeper into the types of tools and types of materials used to make them. Making stone tools by chipping pieces away is called **flint knapping**. There are still people who know how to do flint knapping that can come into the classroom for demonstrations.

If you don't speak of fire the first presentation, bring it up later. It gave great benefit to human society. With fire, humans were provided food and warmth. Fire scared animals away. It may have extended the time people could stay together at night and stay awake to tell each other stories. We don't know who or how someone first began to make use of fire. But sometime or other, someone did. We again can feel gratitude to that person or persons who figured out how to use fire. We should be aware that **in controlling fire, human beings made a scientific chemical discovery, a discovery about chemical change.**

These people, these **homo homo-sapiens**, had working intellects. We need to give them their due recognition for their achievements. We should not downgrade them or their intelligence with the children.

Later on in the chart, we see **arrowheads and other new tools**. That's another scientific fact – that of making these new tools. Which stones could be shaped

into the form needed? If I want to make a nice sharp edge and I am hitting it with some larger rock, I need to figure out how and where to hit the stone to flake it in the right way.

Around 23,000 BC, we have tools made out of bone and antlers. People used some tools for scraping skins. We can see this on the chart. Then someone must have had the idea of using the skin for their own covering – for clothing. Think of the trial and error, the striving for perfection, that must have gone into the making of clothing.

We have a hint of some of the animals on the timeline. And some of the children will want to go into some of those. Which ones are still alive and which died out and when? How big were they? Going out to a museum is a good complement to these investigations.

Also, people must have been working with a language in order to cooperate. These people pre-date the time of Proto-Indo-European language.

**This timeline is used across multiple presentations centered around the gathering of human beings and their fundamental needs.** There are more presentations that can be developed by picking up on the interest of the children and which lead to new interests.

So we are proceeding with the story of human beings. We are still a long long way from us, today – from where our story, our part – comes in. Leave this timeline down as long as there are children looking at it. Ask that the last person roll it up – showing how to roll it from the end, not the beginning. **A few days later, not too long, get it back out and discuss new aspects of it.**

From the teacher's standpoint, we are trying to trace these creatures called human beings and to show they have a place in the universe – both as a group and also as individuals – just as each individual had their part to play in this timeline. We don't know their names but certainly they had a job to do. This is why human beings are important. It is why we have a responsibility we can discuss with the children. We can help the children to see that we have a choice in life to be responsible to each other and the world or not. We have a gift and a responsibility. This is how music, art, physics, chemistry, language – all of it – will fit in. This is what will ultimately help human beings in the future.

## Fundamental Needs of Human Beings

When we began history work, we saw the two fundamental needs of human beings. We mentioned the **material and spiritual territories** having been built up by humanity. In our history stories, we first show the child how the needs of human beings had to be satisfied physically to keep the species alive. Human beings had to have food and shelter to stay alive. But human beings are not just a physical body requiring food clothing and shelter.

We also have **intellect and love**. This means they also have needs to be met – needs of the mind and spirit. And it **seems as if these needs are never completely satisfied**. The intellect is always searching, trying to understand more or to understand something more clearly. It's never at rest.

To begin with, the **child is made aware of the environment that had to be prepared before human beings could come. So we have the creation of the earth** and the immense length of time that it took for this to happen – in relation to the time that human beings have been on earth. We give these impressions to the children with symbolic charts and timelines. In particular, we show the immense length of time using the **black strip**.

Then we take up the **furnishing of the earth – the plants and animals. We show the work of these in the cosmic plan of development**. This made it possible for human beings to come to an environment in which they could live and from which they could make greater possibilities for their lives. It's the environment of the earth, the sun, moon and stars, the air and water, the plants and animals, which all provide the means for human beings to satisfy their needs.

We can't give an exact date to the different stages in the creation of the earth or to the development of plants and animals or to the coming of human beings. Whatever dates are assigned one year, are readjusted with subsequent discoveries. The charts and the timelines we use with the children are illustrative material for this period of natural history. They're impressionistic and symbolic only. Even when we come to the coming of human beings and to the early stages of their life on earth, these can only be given impressionistically since it is the time of pre-history and we don't have written records. The **only evidence we have are the few artifacts that have been found and the few remains of early human beings that have been found**. From that little bit of evidence, we must piece together what we think happened and what we think it was like.

As we piece together these stories for the children, we want to help them understand that human beings first had to satisfy their physical needs in order to go on existing. They had to find **food**. They had to find **shelter** – since they probably arrived during an ice age when it was cold on earth. They had to have developed **clothing** to protect their bodies. They needed **defense and transportation**.

The spiritual needs were there also right from the beginning but many people think the spiritual needs couldn't really surface until the material needs had been met. Then there was a little extra time in daily life to pursue those spiritual needs. Not everyone thinks this; some feel all needs were felt at the same time. Ultimately it is the child who decides for himself.

When we begin to deal with these needs with the children, we want to help them first of all understand that in the beginning, human beings used what was right around them in the immediate environment to satisfy their needs. They were hungry and so looked right around where they lived – to find what they could eat. It began to rain and so they wanted to take shelter – immediately. We can imagine that they used whatever they could find right around them; caves, rocks cliffs and so on. Their first means of transportation was their feet and legs – what they came with.

As time went on and **human beings began to unite together into groups** to form societies, we can present to the children the various ways in which different societies found solutions to the problems connected with satisfying fundamental needs. **One way of presenting history to the children is to take one of the fundamental needs and to see where it can lead to different kinds of explorations.** For example, if we take the need for food, always a popular choice, we can ask, “What foods do we have today where the name of the food indicates where the food originated.” Along with that goes, “Which foods have a name that sound like it comes from a certain place but didn't originate there.” “Who named French Fries”? This kind of question can be thrown out to the children for discussion and exploration.

Another avenue for exploration is to look at certain foods such as oranges. **How come we can now find oranges growing in our country when they did not originate here? How did they get here? Who brought them here and when?**

Another approach is that the same sort of study can lead to an exploration of how a food was discovered. What exploration developed related to foods to make those foods more readily available to other people? The history of our country comes in here too. What was Columbus looking for when he bumped into us. He was looking for spices! What foods were discovered here that they didn't have in Europe? Corn, chocolate and other items were taken back to Europe by the early explorers.

So you can **pick a food, look for the geographical region in which it originated, identify the conditions necessary for it to grow – and then look at where it has spread to around the world** via human beings. This can be done not only related to food, but also to clothing – taking the silkworm for example. **The silkworm originated in China and were brought here. They all died because they were placed on the wrong type of trees. They had to bring the silkworms as well as their preferred tree before people successfully transplanted silkworms.** History is much more interesting in this kind of inquiry.

As time went on, the satisfaction of the needs led to the building up of what we refer to as the physical and spiritual territories - a particular cultural environment, which became the property of a particular group of people. As these groups became

identified, they took on a name. Gradually, one group became differentiated from other groups. This gave rise to psychological confidence in group-identification but also to tribalism.

Another area of study for the children, which can again start off from something like the study of satisfying our need for food, is **the different ways people become aware of how other people are satisfying *their* needs**. How did they begin to find out what others were doing? **People came in contact with one another by various means and any or all of these ways can be studied with the children.**

One way is by **peaceful penetration** from one territory to another. Throughout history, where do we find that happening? There are instances of this in pursuit of a place to settle – immigration. And **simply spreading out** over geographical regions led to this also. This invariably is motivated by some need or other.

A **third way in which people come in contact with each other is through war**. Throughout history, we can cite examples of how people have come into contact with each other through warfare and ultimately learned aspects of each others way of life. Sometimes **people are forced out of their territory** into another territory. Australia is a case in point, as is the United States. We look at where people started, where they ended up and the result of that dispersal of people. **Slavery is an example of forced dispersal.**

Another way **people mix is through trade** and commerce. There are any number of possibilities of examples. What has been the result of the contact through trade? **What have people learned from each other?** What early nations traded with each other and how did they pay for goods? What happened to the economies of those countries? What transportation did they use? What is the nature of trade today? This work **leads to our later study of economic geography.**

Another way in which contact between people comes about is through **exploration**. Who, what, when, where – are all categories of questioning that drives the child's exploration.

Another way of dealing with history related to satisfaction of needs is the study of **the different ways in which human groups have come together** and developed. First of all, we have the nomads and their two subdivisions. There are the **hunters and food-gatherers**. And then there are the nomads who are **pastoral people**. Another group to study are the agricultural people who satisfied their needs in a different way from the nomads. **Third, there are the urban dwellers.**

Again, using food as the topic of study, the children can learn about the **two types of nomadic peoples: the hunters and food-gatherers** followed the animals they hunted for their food. They tended to live on berries and grains as they traveled in between hunting. They are the people who likely tamed the dog and began domestication of animals.

The **second group of nomadic people are the pastoral people**. It was these people who eventually came to realize that there are different seasons to a year. They realized that during a certain season, the grass grew – and if the grass was eaten by the

animals, it would grow again. So, **if they brought the animals back to the same spot, there'd be grass there again for them to eat. It's from this realization that came the different types of nomadic life – a life where the people didn't follow the animals but rather gathered them together and took them from place to place.**

Developing from this study can grow the study of the beginnings of agriculture which developed from the fact that nomads traveled from one grass-growing region to another. They returned to a certain place where they found the grass growing again. And some of them realized that amongst the grasses were also grains which could provide food for the human beings. **Once they discovered these grain-producing grasses, as time went on, they finally realized that they did not have to continue the hunting life-style; they could grow grasses and grains and stay put in one place.** And so we have the beginnings of agriculture.

One thing to be careful about with the children is to not give the impression that when agriculture started, nomadic life disappeared. The two ways of life both went on and still exist today. Some people became agriculturists and other continued living a nomadic life.

In this way, we trace human existence from early on to the present day. Where are there nomads today. What is their life like? Where are the agriculturists and what is their life like? And the urban dwellers – what is their life like?

Another discovery that came from the people who had settled down - in order to grow, plants need water. And if that water didn't come from the sky, humans had to provide it or else the plant life would die. From the agriculturists come the beginnings of irrigation. Early agricultural societies were successful near natural sources of water. There was the possibility of irrigation there.

Then came another development. As settled agricultural life went on, the needs of providing shelter and roads for travel within the territory led to the realization that some of these jobs could be divided up and shared out to different people. So, one person didn't have to do everything for himself. This brought about the **organization of society through the division of work.** We get the beginnings of the construction workers, the scribes, the priests and agricultural workers. This is what eventually led to the development of urban organization. Developing from the settled agriculturalists, the urban organization came about in more than one place. And it led to what are called the Four River Civilizations – and to development of each of those cultures.

The division of work and the organization of society led to the making of laws for these organizations. Hierarchical groups developed and punishments were developed for those who did not keep the laws. The laws differed from one group to another. The laws that a nomadic group needed were different from those needed by an urban group. So the laws that developed had to suit the needs of the group. As the pastoral people came in contact with the more settled people, they began to exchange products and they developed systems of bartering. All of those can be looked into.

As the children study the different changes within society, they will also

begin to study the geographical changes of land and water forms, and the accidents of nature and climate which have caused once fertile land to become arid land. Where has this happened? The Sahara is a great example of this; it was lush at one time. A change like this, brought about movements of people who had to look for new land where they could meet their needs. Sometimes, wars resulted as people fought against each other for the same territory.

So we have people living different kinds of lives. The **nomadic people became physically stronger than those living a more settled way of life**. We can imagine the wandering lifestyle had many hardships and those people had to be hearty and strong or they didn't survive. They became more resistant to fatigue and climatic changes. So it became **easier for them to conquer some of the settled societies**. Remember, the nomadic people probably first tamed the dog and domesticated animals – so many positive things came from this way of life.

It was the **nomads who first used animals for transportation** and brought about the idea of discipline, within a group and banded together for a common goal. But these nomads became physically stronger and so some of them such as the **Hittites** were able to conquer other people. **The Hittites were a nomadic horde. They had discovered the use of iron which they used to make tires, other kinds of wheels and weapons**. They **descended upon the Egyptians** and conquered them. The Egyptians only had wooden wheels – obviously not as strong. The Hittites equipment was more hearty as well.

Another group of people called the Hyksos descended upon the Egyptian army in about 1700 BC and because of their greater strength and more powerful weapons, they were able to occupy and rule Egypt for many years. The “shepherd kings” in Egyptian history are from this group of Hyksos. The children can explore one or two examples of nomadic people who were able to take over a more settled people due to their easier lifestyle – leaving them easier to conquer.

Another study of the need for food can lead to the reasons behind explorations that human beings have undertaken. In earliest times of human beings, explorations came about through the possibility of trading with each other. The early Egyptians explored the Nile river. The early Phoenicians explored the Mediterranean. It was early explorations like these which opened up territories and countries, making it possible for other people to use products and other artifacts of those cultures.

As time went on, the means to satisfy the needs of human beings have increased and they continue to increase. We can begin to see the spiritual needs which eventually led to what are sometimes thought of as abstract explorations. These are the explorations of the mind and spirit. It is these kinds of explorations that pushed Scott to Antarctica and that drove Hunt to climb Mount Everest – since no one else had done these things. Human beings have this need to know, to extend their field of knowledge and find out what cannot be seen with the naked eye. And they have the need to pass this information on to others.

We've spoken of taking up the needs of human beings as a basis for studying

history. This type of study starting off from these ideas, can go either vertically or horizontally. Traditionally in history, the way to go is vertically; meaning, from the beginning in a certain place or region and chronologically tracing developments.

**Horizontal study means that there are no dates needed but another theme provides structure – such as hunter / gatherers.** We can look at the hunters and food gatherers where we have evidence of this first starting. In what parts of the world have evidence of this been found? What animals were hunted and what food was gathered? What kind of life did they build up and what was their culture like?

Or we could take the pastoral people and see which of the hunting people became pastoral – and which ones resisted that change. We could take the agriculturists and look at where we have evidence of agriculture beginning. What kind of farming took place? What conditions made it possible for agriculture to develop in those places? Why did urban swelling come about where it did? Did it follow naturally from the agricultural revolution or did it develop on its own independent of agriculture?

**With a horizontal study, one arrives at a comparative treatment; it is a comparative study. With a vertical study one gets a chronological treatment. Our history studies should include both types.** But especially when we consider early human beings, it is going to be horizontal since we don't have dates to apply.

As we **come to the time of written records with the children, we come to the point when we need to make sure the children have a concept of chronological time.** With first plane children, we said it's only near the end of the first plane that children can really grasp the concept of time. Certainly, the clock and telling time on an analog clock can be introduced in the primary classroom. But often we need to work on telling time in the elementary classroom as well. We need to make sure the children can read a clock, understand a calendar, know that days follow one another, understand there are seven days in a week, weeks make months and months make up a year. The children should have been introduced to the calendar in the primary classroom as well. They can often become aware of the passing of days as they follow the calendar.

In the early history work in the elementary classroom, the Timeline of Life, First Timeline of Human Beings – we don't make references to dates. It's hard for us to grasp billions or years and so we also must give the children a chance to build up an understanding of that kind of time. They also need to understand that not everything is in the here and now; this is why we use the Black Strip and Hand Timeline – to demonstrate how long ago all of this began.

As dated timelines become necessary and we continue to explore human beings, we need to make sure the children understand this chronological passage of time. We'll **help the children build up a chronological picture of societies that resulted from meeting their fundamental needs in different ways.** It's important that we **present these things in the context of cosmic education – not as a separate subject of history.** We want to help the child realize the accomplishments of all humanity and to realize the debt that is owed to unknown human beings who came before us.

Throughout history, societies have come about and eventually turned into

nations through finding ways to meet physical and spiritual needs. These ways have formed bonds amongst the people of a region and have enabled them to recognize themselves as members of a group. The fundamental needs have resulted in the common likes and dislikes that a group has as well as the common way of dressing, speaking, beliefs, literature and art – everything that made up early Egyptian, Chinese, Mayan and other cultures of the world.

We as teachers must understand that as we present history, we're presenting the ways in which a group has built up their way of life and what it is that distinguishes one from the other. All of this is a result of satisfying the same basic needs that all human beings have.

We should **realize there is not just one set presentation for these charts.** They can be used in a wide variety of different ways. Since we are human beings, they are a piece of material that **can be used very early** – related to how we meet some of these needs. **It is easy for them to understand we all need the same things.** On these charts, it is easy to see different ways of meeting needs. We can ask, “How do you meet your need for transportation”? We might suggest they make their own charts for how they meet their own needs.

We can also use these charts with the Timeline of Human Beings. You could bring out that timeline and this first fundamental needs chart, pick one of the needs, food for example, and ask what examples we see on that timeline for how human beings met their need for food. This can be the focus of exploration for that timeline. What about housing and shelter? What example do we see on the timeline for how human beings have met that need? The same can be done using the second Timeline of Human Beings as well.

These charts can also be used relative to any group of people we introduce to the children. We'll **talk about civilizations, the Mayans for example. We could use the spiritual / material needs chart and discuss how the Mayans approached meeting their needs. This helps them to picture what their lives must have been like.**

## History Question Charts

We approach history in its broadest sense; normally what we call history is studied in anthropology, archeology and so on.

**As we talk about human beings, we speak of founding the science of discovery, the science of invention.** In the course of building up their lives, these people have developed what we call the material and spiritual territories.

These territories have been built up in all different places around the world – **based on what the people found in their own environments.**

It's not necessarily the most well-known people from history that made the greatest contributions to our everyday lives. Often, it is the unknown creators and inventors who developed things which have been passed down to us today and to whom we owe such a debt of gratitude. We want to help the children understand this.

In a book called “Man Makes Himself”, by Professor Gordon Childe, the author traces development of human beings from their first **ability to make a fire and to fashion simple stone tools.** These two discoveries helped humanity survive amidst the animals that were also living in those early times. Professor Childe also describes the development of food producers, emergence of citizen states, the rise of foreign trade and the urban revolution. He lists what he considers to be the basic discoveries of human beings and their application to science.

The **discoveries he lists are:** artificial irrigation, the plow, the harnessing of animal power, sailing boats, wheeled vehicles, orchard husbandry, fermentation, the production and use of copper, bricks, the arch, the process of glazing, the seal / stamp used for identification, a solar calendar, writing, numeral notation, bronze, smelting iron, the alphabet, and aqueducts for urban water supply.

Another way in which we can approach history studies is to **take any one of these discoveries and explore where it was made, when and by whom was it made.** How did it effect the lives of those people? Did it spread? Is it still in use today? Has it developed into something else? And we recall this is just one person's list. One could challenge the children: “What do you think the 3 most important recent discoveries have been”? We don't know which might be the most important for recent discoveries since their discovery is so recent.

When we study the sequence of human being's history, we find that, for the most part, pre-historic societies are found represented not just by skeletons that remain and are found, but by house-plans and other artifacts such as pots and pans, personal ornaments, burial and other ritualistic artifacts. These are the remains of human beings that archeologists classify into types. It is when the same type is found together over and over again, but in a limited geographical area, that they're then grouped together and

termed a culture. Cultures therefore stand for societies. Because if similar remains are found over and over again, they must not have been developed by only one individual. They must have been developed by a group of people – a society. And we can imagine that groups need to communicate and cooperate with one another. It is this kind of evidence that is found and examined regarding pre-historic societies. Then archeologists give names to these cultures to distinguish one from the other. How are these names usually derived? Often it depends on the place where artifacts are found.

In developing lessons for the children, so we are able to **give the children the keys they need to investigate and explore history**, Dr. Montessori developed some material that we use. In the early years of her work in Europe, especially in Holland while she was working on material for the older children, she met a Dutch woman named Mrs. Prins-Werker. She helped develop these materials. These materials are called the history question charts.

These **charts serve as the foundation around which our history stories should be developed** and they serve as a tool for the children's work. The material is based on sets of questions – four of them. These questions, first of all, provide the teacher with the guidance for giving the children information and the kind of information that they need for history work. These charts are meant to be used by the teacher first, as we prepare new presentations and as we give them. They can be used for any period of history, for any society, any civilization, any culture or nation.

The teacher would **choose questions from the charts**, where the information she is going to give in answer to them, **builds up for the children, a picture of what that society was like**. What geographical surroundings were like, what the people were like and such things are approached. We can ask how people met their spiritual and material needs in answering these questions. As one develops interest in a group of people, **questions can be asked and answered over many days of study**.

We refer to the principle of limitation. We need to **limit how much we give in our stories** so that we **don't give everything and leave nothing for them to find** on their own. Yet, we need to **give enough information to develop interest**. So, one must find that balance in developing lessons. That gets easier with practice. One guide is – if introducing a culture the children know nothing of, one gives more. This is common sense – if they know more, give less – just enough to build interest.

The answers we give need to be of **high interest but harder to find**. When children go to books or to the museum afterward, we want them to find different information from what was given in the story-presentation. When telling the story, end it in such a way that the children know there is more to find out about in this subject. So we open the door and invite them to find more for themselves. Sometimes we also need to give an indication of where to look in order to find out more. If we know there is an exhibit at the museum, for example, we should mention it.

In order to make use of the **psychological characteristics** of the second plane of development and to arouse interest in the children, we need to show them what **the lives of people were like, what their emotions may have been like, what they did**

**for fun and amusement.** We must show them their daily activities. If we don't bring in these things, the children tend to think it was the same as it is today. Their mental picture has to be filled in. These presentations are stories that the teacher needs to prepare with great care. Often our presentation is accompanied by an artifact or fossil or some other material that fills in the child's mental picture of this time and culture. These aren't the sort of presentations one gives everyday. They must be thought out carefully and planned for – **approached with a certain seriousness** – since there is a great importance to this work.

Mario Montessori used to say is that the **first thing we must do is to prepare the child's mind to receive the lesson.** He said the way to do that is to **tell the children ahead of time** that there will be an introduction to one of these stories. Here is where planning ahead comes in. One might say, **“Next Wednesday at 9 o'clock, I'm going to be telling a special story about the Aztecs.** If you're interested in coming plan to be there Wednesday at 9.” By telling them that, we accomplish a level of eagerness and interest – maybe the children even begin to look into things ahead of time. Curiosity builds up to the point of presentation. Especially for those who've been in your class a while, this will catch on and spread among students in the classroom. They begin to compete for this information. So the exploration should begin even before we give the presentation.

This represents a greater responsibility for teachers – regarding lesson plans. We must **prepare ahead of time** – often over the course of weeks – ultimately over the course of three years. With some of these presentations of different cultures, it can be a time when one could bring in a **guest lecturer.** This obviously requires forethought as well. We normally don't want someone who has no knowledge of Montessori and the ages of children we are dealing with. When there is a guest lecturer, teacher also is informed.

The **History Question Charts consists of four charts.** The charts nest together physically – allowing one to see the title of each at one time.

The **first chart**, which is brown, is entitled **“The Nature of the Country.”** The questions are:

- What were the soil and climate like?
- What were the flora and fauna like?
- What people lived there?
- How did they come?

The **second chart**, which is green, is entitled **“The Practical Activities of the People.”** It has to do with the people themselves. The questions here are:

- What were the types of work and occupations?
- What did they produce?
- What tools and techniques did they have?

- How did they find their country and how did they make it more habitable?
- How did they make use of natural resources?

The **third chart**, which is blue, is entitled “**The Intellectual and Spiritual Aspects of the Culture.**” The questions are:

- What language did they speak?
- What was the education like?
- Who were the educational leaders?
- How advanced was their learning?
- What was their art like?
- What were their ideas about life and death?
- Who if anyone, was the spiritual leader?
- What concept of justice was there?

The **fourth chart**, which is pink, is entitled “**Relations Within the Group and Other Groups.**” The questions here are:

Did they trade among themselves?

Did they trade with others?

Did they barter or use other means?

What about their wars and conquests?

Did they have slaves?

What about travel and migration?

How and why did groups settle where they did?

How were their money and goods held? Individually or by the group?

What about dress, food, housing, customs and family life?

How were they governed?

What care did they give to their children?

These are questions that are common to all groups of people. They apply to all people though the answers may come in different forms.

Presentations can be made horizontally. We do a comparative study – not chronological. Societies living during the same period of time but in different parts of the world. One could ask the same questions about each and compare the answers found. Because the child has a reasoning mind, they begin to compare and contrast the similarities and differences in the spiritual and material territories.

The presentations can also be given vertically. One could start at the beginning of a civilization and follow it chronologically until its decline. Presentations can focus on the questions but can be carried out comparatively or chronologically.

After the teacher has given his story, the children should begin exploring and looking for what else they can find. Perhaps they want to know more information about the same questions so they begin reading books, illustrating or writing about the material – in some way, developing their own part of the project.

If one has appropriate books in the classroom, and we usually need one or two about the group being introduced, make sure to let the children know where to find them. Let the children know which section of the public library will provide more in depth information. We need to be the source for that kind of information – where are the resources to help themselves.

The lessons given by the teacher should provide the children with information that will be used for subsequent work with the various timelines in history. If we don't help the children understand the different conditions of life, difference between societies, they won't really understand events that we speak in those different places and times. The events have to be put into a social structure.

When we prepare a presentation using the charts, we can use one chart and choose questions (one or all) from that chart only – or we could use questions from several charts (one or all) – and the children can use the charts in the same way. They can choose whatever questions they wish and find answers to those questions. So, it is important to realize the charts are not a fixed material to be gone through in a procedure. They are guideline materials – guiding the line of research.

Each chart has a white spot at the end. In our environment, we can have a set of prepared answer cards. This can be developed for any group that one chooses to develop it for, as a model of the sort for answers that are sought. For early human beings, for example, the **answers appear on cards that are then set as a pile on this white area at the chart's end - as each card is taken and read to determine which question is answered – placing the card under the question each time.** Make one set of answer cards to begin with; allow the children to make others sets.

Another time **it is appropriate for answer cards is for required information for the public school curriculum.** If teaching in an upper elementary class, often state history in the US is part of that curriculum. One could make a set of prepared answer cards for that required information. This provides a manipulative activity for them to help learn that subject matter. The same can be done regarding one's national history. **We should always remember that prepared answer cards are used only after the teacher has done the related presentation.** It then provides needed repetition.

January 10, 2005

## Three Phases of History

Adapted from notes kindly shared by Carson Wright

As we present the history of humans to the children, our stories build up an impression of the construction which has gone on which has resulted in the physical and spiritual territories. We try to help the children understand that this process, this life lived in particular societies, has been linked to science and technology for the **material territory**, and to the development of customs and ideals for the **spiritual territories**. Within the long span of human history, we also have to show the children that there has come about from time to time certain innovations, which have brought about a revolution in ways of life.

And generally the result of these innovations has been an increase in population; these revolutions mark off stages in the process of the development of societies, especially those which became nations.

### ***Hunter Gatherers of Paleolithic (Old Stone) Age and Later***

First of all human beings **arrived on earth, perhaps a million or 2 million years** ago - it is really hard to say. We do not know a great deal about what their life was like or how they lived from day to day. What we do know is that early humans began life as food gatherers and they caught food which they could catch around them and what was edible. This food gathering economy lasted through the **Paleolithic or Old Stone Age** as the archeologists call it or as the Pleistocene by the geologists.

A food gathering economy is still practiced by some peoples (Borneo Malaysia, bush men in South Africa). So, it is important that the children don't think of the old stone age as ending on a particular date, and then the next age starting and going on from there, what is termed the old stone age closed in certain parts of the world and not always at the same time. We need to be very aware and to inform the children - these changes have been very gradual and varied considerably.

At the time of the **Upper Paleolithic there were groups in Southern France and North Spain** who lived by hunting and fishing and gathering fruits and berries. These cultures have names, such as the **Solutrean culture and the Magdalenian culture**; these are the people who produced the **cave art in Lascaux and Altamira**. They lived from about **35 thousand and declined about 10 thousand BC**. You will see

them mentioned on the next time line that you will see soon.

## *Agricultural Revolution and the Neolithic Period*

**Between 10,000 and 12,000 years ago** comes the stage when we begin to know a little bit more about the history of human beings, some societies, first in the near east - **started to cultivate plants and breed animals**. In this way they were able to greatly increase the food supply. This **food-producing agricultural economy** came about during what archeologists term the **New Stone Age**. It is this change from a food gathering economy to a settled agricultural way of life that is known as the **agricultural revolution**. But this *does not correspond with the economic sense of a particular time* because the **Maoris in New Zealand were still Neolithic in 1800 A.D.**

The **beginning of the domestication of animals and the cultivation of grain first took place in the ancient near east about 10,000 years ago in the area called the Fertile Crescent**; this is the area from Egypt through Palestine to Mesopotamia and western Iran. At this time this was a land of wild wheat and barley. , wild sheep and goats. In the book The Ascent of Man, Jacob Bronowski says on p.64, “The largest single step in the ascent of man is the change from nomad to village agriculture.” One of the things he says is that a happy conjunction of natural created agriculture, he says in the old world this happened about 10000 years ago, but he goes on to say it happened more than once. Almost certainly different groups independently of each other developed agriculture.

One of the very interesting thing is that both bread wheat and maize need human beings today to propagate it, regarding bread wheat these wild grasses had seeds that flew in the air, they could very easily reproduce themselves. At some point when wild wheat cross-pollinate with a natural goat grass became Emmer. It could still be spread around with the assistance of the wind, then something else happened. Emmer cross-pollinated with another goat grass which produces bread wheat. In its most primitive form the ears were more open and if they broke open the seeds could still scatter. But with bread wheat this ability became lost, and this is why bread wheat needs human being to harvest it had spread the seeds to grow. (More importantly **humans could control the particular traits they desired**) The same thing is true of maize. For good information in the classroom, include the book called “Corn is Maize”, by Alik.

In order to harvest the wheat these early farmers developed a sickle with a serrated edge to cut the stalks down. It was important, because it allowed the seeds to stay in place when the stalk could be gently sawed. This shows how science and technology had to progress for agriculture to progress.

**Early farmers used fire**, especially in Neolithic times. Large forests in temperate Europe were cleared by stone axes, which helped to clear the forests of its trees with the assistance of fire. This allowed for agriculture to begin. The stone blade or a shaft created a hoe, which allowed for the turning up of the soil, plowing then developed. Bronowski says on page 74, “ the most powerful invention in the whole of agriculture is of course the plow. We think of the plow as a wedge dividing the soil and the wedge is an important early mechanical invention, but the **plow is also something more fundamental; it is a lever**, which lifts the soil and it is among the first application or principles of the lever.”

The move beyond simple agriculture in the Middle East was made possible by the **domestication of draft animals**. Bronowski say that the failure to domesticate animals in the new world was limited by the lack of domesticated animals. (It wasn't as much a failure as a lack of domestic-able animals; in North and South America, the only available animals were the Lama and the Alpaca).

## ***Urban Revolution and the Rise of Cities in Late Neolithic Period***

At the **end of the Neolithic period**, **two major ways of life** existed. There were **hunters-gathers, and settled farming life-styles**. No town had more than a few hundred people.

The **next economic change**, came with the transformation of some of the **river side villages into cities**, about **5000 years ago** in the valley of the **Tigris**, at the **Euphrates, Nile, Indus** and then a little bit later, at the **Yellow River in China**. The change came about when the farmers in the agricultural areas created a **surplus of food**; they produced more food than there families needed.

They could give their extra food to others, which **allowed those groups to take on specialized tasks** or labor. This led to the development of crafts and, priests, clerks, ruling officials. This economic change is known as the urban revolution, it is brought about city dwellers.

This **Urban revolution** first took place in southern Mesopotamia; it develops on the flood plane on the Tigris and Euphrates, Bronowski remarks, “Civilization can never grow on the move.” In other words, nomadic people who moved

from one place to another could not stay long enough to develop a civilization. In order to make civilization possible, people had to have settled down with a **society able to live cooperatively** together. Life had to be able to be organized in such a way in that there is a distribution of labor, then came the leisure time to develop some of the facets that make up civilization. This is shown where there is an advanced society who are proficient in the arts and sciences and there is a certain social proficiency. **Life generally centered around the temple.**

## ***The Metal Ages:                      Bronze and Iron Ages***

There are **two separate stages** in the early stage **of civilization**. The first **2000 years** corresponds to what archeologist call the **Bronze Age**, as copper and bronze were the only metals used to make **tools and weapons**. The second stage began about **1300 BC**, when an economical method brought about the **creation of wrought iron**, this age was called the early **Iron Age**. Think about what a difference this made, stronger, weapons and iron products. In the near east, at about the same time came the invention of the alphabetical script a much better way of keeping records after **700 BC retail transaction** could take place when **coined money** was used. Coins were first used probably **in Asia Minor**.

## ***Industrial Revolution***

The first stage in the next major world change happened in Medieval Europe and was called **Feudalism**. It essentially involved a person, called a vassal, being given the right to develop and profit from someone else's land in **exchange for his allegiance and service to the landowner**. This system in affect tied an individual down to a particular land, prior to that the people were semi nomadic, moving around planting where ever they wanted. With Feudalism though, the agriculturalists stayed in one spot, this provided for a more fertile land.

With the opening up of trade with the Americas the **lands closest to the water thrived**. As industry developed mass produced goods became available, power tools, and the introduction of the factory, all of these goods which were produced in the factory and the food products which became increasingly available were shipped to those societies which boarded the sea coast. This began to be known as the first stage of the **industrial revolution**.

The children need to be made aware of these stages in the history of human being until the present day, each day lasted for a long, long time. And each stay produced

its own science and its own economy, in the classroom, with the children, we need to relate these changes to a particular society or two. What other examples do we know of? With the older children you can say that people think we are in the midst of another revolution?

January 11, 2005

## **Second Timeline of Human Beings**

### ***Orientation to Timeline***

The first thing we notice is that there's an overlap with the First Timeline of Human Beings. The first timeline starts about 500,000 BC and goes to 1 AD. So it covers the Lower Paleolithic era.

The Second Timeline of Human Beings starts about 23,000 BC and goes to about 2100 AD. Again we have pictures that are sparsely placed as well as some writing. Right in the beginning we see ice representing ice ages. And the blue line is cold – the red line is warm. What do you notice about the temperature over time. It's warming up!

Also notice that we have vegetation – and words, Tundra, then Steppe. We use the words Steppe for grasslands in Eastern Europe and Asia and called the Prairie in the Americas. Another type of vegetation – we see agriculture.

We also have Upper Perigordian cultures, the Solutrean culture and the Magdalenian cultures. These are people identifiable as cultures because of the tools and clothing we have found distinguishes them enough to give each group a name.

We see hunting, people herding horses over a cliff for food, someone working to make a tool, a woman sewing and we see domestication of animals too. We see house-building, a Stonehenge like structure for an enclosure. We see someone weaving here. Finally, we see some art objects in the shape of animals.

We also see at the time of recorded history the terms agriculture, metal, Great Pyramids, Trojan War. This can lead us into discussion with the children. Who built the pyramids? Who fought the Trojan War? This line here says cave-beasts die out – it looks a little confusing; they were Sabretooth tigers and some types of bears.

### ***Story Presentation***

(gather a small group; begin with timeline rolled up)

I've noticed that some of you have really looked at that first timeline of human beings. I'll bet some of you have thought about the special gifts that human beings have. Can you remember what those are? (reason-intellect, love-will).

And we know that with those tools, people lived their lives. They got food,

clothing, housing – let's take a look at some more of this story. I have a second timeline of human beings.

(unroll entire timeline before the children)

What do you notice? What interests you first about this timeline. (the lady sewing). Yes, do you remember on the end of the first timeline, there was a tiny picture of a needle. Before this, people have to poke a hole in the skin and then poke their sinew (tendon or intestinal material) through the hole – kind of like lacing shoes. But at some point, someone figured out that if you took a little bone and put a hole in one end and sharpened the other end, you could poke the hole and put it through the skin all at once. So they could make clothing more easily. What else do we see. (man plowing)

We can see this man is using oxen to plow, can't we. Before this, he did this work by hand. I know some of you are interested in plants. Do you notice the Tundra that is growing. All this occurred during the Upper Paleolithic.

(Discuss different between Upper and Lower Paleolithic)

(tell my own story here)

Let's look at this word Solutrean. See this picture here (point) of this man. The Solutrean people got to be very good, better than everyone before them, at making stone tools. They not only knew how to pick just the right stone to make a sharp edge, but they knew how to break the stone to make exactly the right size for a spear or arrow head. And they knew how to work it into exactly the right shape.

After the Solutrean people came the Magdalenian people. They were even more brilliant culture; that was their reputation. I wonder how they got that reputation? The Solutrean and Magdalenian people are also responsible for much of the beautiful cave paintings found at Altamira and Lascaux. But that's a story for another day. But we have books on the shelf you might want to look ahead to – on your own ahead of time.

The last person finished with the timeline – please roll it up, beginning from the end. Don't forget to ask me for more stories if you have interest in anything you see there.

## **Epilogue**

This is a timeline you bring out over and over again. You can pick up on different aspects each time. It lends itself to use with the History Question Charts. You take a question, look at the timeline and develop a story around it. The children can use the History Question Charts to guide their own explorations.

This timeline can also be used in conjunction with the Fundamental Needs of Human Beings Chart. They could trace one need across period of time. Hunter-

gatherers include the Solutreans, Magdalenians – but toward the end of the chart we find agriculturists. Keep the story conversational or you can direct the story.

We have books on the shelf. “Life in the Old Stone Age” - part of the Cambridge History Book Series – is a good choice.

**For oral exams, prepare a story to tell about some aspect of this timeline. As we practice, become more familiar with these pictures – we need intelligent aspects to add to what is visible on the timeline.**

### ***Sample for Further Presentations***

(using History Question Chart, Second Timeline of Human Begins unrolled and small group – this is a later presentation than the first presentation of the Second Timeline of Human Beings)

Today, I want to take a look at about this time (point to the end of Neolithic Period and Age of Metal). This is from about 8000 BC to about 6000BC. I've got my History Question Charts too. I want to talk about this question from this chart – The Practical Activities of the People. The question is, “What tools and techniques did they have”?

During this time, people discovered something quite amazing. We know that for years and years people made tools from stone. They could make very sharp spear-points and arrow-points and knives and axes too. There were a few problems with those stone tools; one was that they broke easily if dropped. Another problem was if you didn't live near where you could find that stone, it was hard to get.

At some point, people found not rock but a metal – actually there was two metals. One was gold and other was copper. Gold is very hard to find so mostly, they used copper. At first what they did was to take a piece of copper, put it on top of a hard flat rock, and then pound it to flatten it out. They'd pound the edges and this way they could make a spear-point or arrow-head. If they pounded it nice and thin, it would be very sharp. And so they used it in this natural way for a very long time.

But somewhere along the way in Eastern Asia, people discovered that you could melt copper and pour it into a mold. When it cooled, it kept that new shape. So, they could make points for spears and arrows and axes this way; they even made jewelry this way. There was just one problem with these copper points. They got dull very quickly. They did not stay sharp – and no one wants to go hunting with a dull spear.

So someone discovered that if you mixed two metals together, melting them and then pouring them together into a mold – they would stay sharp much longer. It was much harder that way. They called this new metal **bronze**. It was a solution to the

problem of keeping a sharp edge on tools. This solution began the age of metal and it changed history forever.

That's my story for today. Did you notice I didn't say what two metals make bronze. I wonder how they made the molds. If you find this information, let me know. Don't forget there are also all these other questions on the History Question Charts to look into. You can ask me for more stories and we can share what we find out.

## **Epilogue**

“Prehistory: The World of Early Man” – edited by Jean Guilaine is an excellent source. “The Wonderful Prehistory in Perigord” (ISBN 2737303818) is another very good source of follow-up.

January 11, 2005

## The Migration Charts

The **migration charts are an impressionistic piece of material** to be used whenever relevant along with our presentations and stories to the children. They are not a piece of material that one would introduce for their own sake – like we might do for the Second Timeline of Human Beings. The **migration charts are a tool more like the History Question Charts and Fundamental Needs charts – tools that accompany other presentations.**

We've been talking about history developing through the ages and some of the ways in which people have built up societies and cultures. We've mentioned that various groups have come into contact with each other in different ways - and some of the various developments that took place as a result.

Sometimes people come in contact through peaceful means and at other times not so peacefully. Both aspects fall into our history stories.

When we're discussing with the children, some of these movements of people, and some of the different reasons that have caused people to move from one place to another, we can sometimes use one of the migration charts to illustrate the kind of movement we speak of.

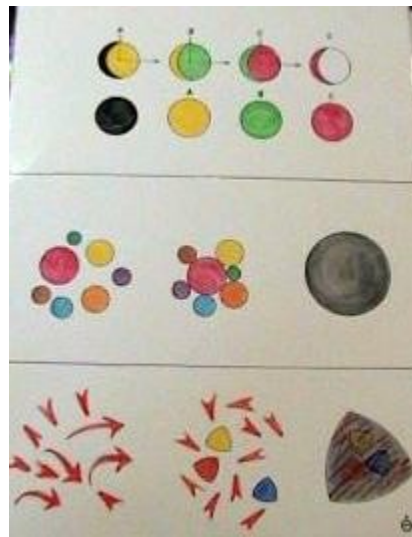
As groups and societies throughout time began to have the chance to **develop a higher culture, a better and more relaxed way of life, some of them developed a surplus of goods.** They didn't have to live hand-to-mouth any longer - as they learned how to use the environment more successfully and to get more from the environment for their lives. The obvious result was a richer and more luxurious society. People were able to live more comfortably.

This form of life depended on a settled habitation, a settled environment. These were the settled-down people and as time went on they were sometimes **visited by nomadic groups** – those who were still on the move. Sometimes **traders from other groups** came in-contact with some of these richer societies and word began to spread amongst some of the other inhabited areas that there was a richer lifestyle in another land. Sometimes this richer land was appealing to these other groups, maybe those on the fringes felt they wanted such wealth and comfort too. Their living conditions were still difficult.

So some groups in some places began to move in amongst a richer group and they found places to settle there. In some cases, **their population grew as a result too. Eventually their population surrounded and outnumbered the original people** who lived there. Then the problem would arise as to whether it was possible to **fuse the two groups** together – or – would the **first group in possession of the land try to fight off the intruders.** As the children study this problem in history, they will come across

many instances where the people began by fighting and ended up peacefully fusing together later.

This kind of movement of people is called *Infiltration and Fusion* migration. It is at the **bottom of the “3-in-1” chart (chart H7)**. The **red arrows represent the group that is moving in. In the middle caption, the curvilinear triangles represent the people already there that can end up surrounded and outnumbered. The groups become blended together in the third caption** but each group might retain some of its original customs, culture and way of living and thinking. They aren't just all blended into one – there is still some distinction between the groups. That's one kind of migration – movement of people.



There's another movement of people. That is on the **top of chart H7**. The top row (4 spheres) of this section represents the movement and the second row is the result. In this kind of movement, there is a **group already living in one place – that would be the yellow sphere in the top row. But along comes another group on the move – symbolized by the black here and this group comes into the territory where the yellow group is living – and displaces them**, pushes them out. So now the black group is living there – as we see in the second row of spheres – the all black sphere.

The yellow group moves on to a place where the green group is living – here in our first row, second and third spheres. They yellow group pushes out the green group. This kind of displacement can happen once, twice or repeatedly. Sometimes a group can keep getting pushed out – just like balls get pushed on a billiard table. That's why this kind of migration is called the **“Billiard Ball Migration.”** That's what we see on the top of our chart.

And then we see in the **middle of chart H7**, a group on the move once again. This time they are a strong powerful group – the largest circle, all red. In some

cases, this larger group moves into territory **occupied by smaller groups and the smaller groups try to live peacefully with the larger group. They don't want to join in with the first group – they just want to co-exist** and live their normal lives. But they become pressured by this larger nomadic group and often times this **Nomadic Horde** – which is what kind of migration this is – starts to draw in the other people. They are drawn in to the group on the move. So these **smaller groups join in with the army of fighters and they become larger and larger and more and more powerful.**

As they move on, they **become one group**, the far right black circle at the right end of the middle of our chart. Genghis Khan and his Mongol horde is an example of this. We spoke of hordes in the development of our language. The **Great Wall of China was built to keep out the invading hordes of that day.**

So those are three types of migrations but there are others.

There is another which is called ***Breaking the Wall (chart H10)***. Settled people began to develop nations. In some parts of the world, some of these societies grew bigger and bigger societies – like some of the ancient empires that held vast lands around the world – such as the Roman Empire. If one looks at the holdings of that empire, we see it was vast. And they exerted great authority over these lands that they held.

Many of these lands were far away from the seat of power. They developed huge armies and they counted on those armies to protect the far-off reaches of land from invaders. Sometimes the people in the center of power built walls to protect themselves. The Great Wall of China is one such wall. Large sections of their territory like Hadrian's Wall in England is another example. Sometimes they built walls around their central city such as the walls that were built around Rome.



At times, the people living in these central cities became more comfortable. Their **lives became easier and they didn't worry about protecting themselves** since they thought this was being taken care of. They counted on those armies. As a result, they

became less able to defend themselves and there came a certain moment within some of the ancient empires when the armies on the fringes realized that they, in fact, had the power. And so in many instances, this led to an insurrection – a rising up against the empire. And a conquest against the central power often took place. This chart (Breaking the Wall) could illustrate the sacking of Rome or Alexander the Great invading Athens.

There is also a migration chart called “*Clearing the Forest (chart H6).*” This type of movement of people occurred where people had moved into a forested area, cleared all of the trees, prepared the land for farming and they farmed the land there. After a certain number of years, planting the same crops over and over again on the same land caused the soil to dry out and lose its nutrients. So, the people would pick up and move to another spot and repeat the whole process.



The brown patches on the chart represent the worn out land. If the same crops are planted over and over again in the same soil, soon the crop no longer grows well. So they had to move on again and again to the green areas. Certainly this happened in settling the **Americas and parts of Europe**.

There are other types of migrations besides those we've illustrated on these migration charts. We could speak about **Sea-borne Migration**, people moving by water across the sea or ocean and setting on another shore. We could develop a chart to depict this – carrying the **Vikings to England** – or early settlers to North America. What are some of the reasons for this type of migration?

Sometimes it was for **lack - of food or religious freedom** – those things that are fundamental needs. So there are many stories possible with this type of migration. **Forced slavery** is another type of migration that has impacted many countries – certainly our own. This is another possible chart that could be constructed in a chart and use to illustrate that history.

The migration charts are to be used to give the children and impression of some of these ideas about the different ways in which people have moved from one territory to another. As we tell these stories and present these aspects of history, we want to try to present a balanced outlook whenever possible. For example, when we speak of the warring people of history – it's easy to speak of the negatives of warfare. But sometimes these were the more adventurous people who brought these characteristics into the region they came into. Our job is to present the picture and to let the children develop their own value-judgments about those things on their own.

These **charts should and can be used to accompany the stories we develop along with the History Question Charts**. Often we can use one or two questions to get started and then use one or two of these charts to illustrate our story as we go along.

### ***Summary of Major Types of Migration***

- Forced Slavery
- Billiard Ball
- Infiltration and Fusion
- Nomadic Horde
- Clearing the Forest
- Migration by Water
- Breaking the Wall

A common reason for migration relates to a lacking felt among a population (such as lack food or lack of religious freedom).

January 12, 2005

## The four Ancient River Civilizations

Thus far we've looked at the **fundamental needs of human beings that provide the framework to the study of history**. We encountered the **History Question charts which guide the students in building up a picture** of a group of people and guide us, the teachers, in creating the stories we tell. We talked about the **3 phases of history** and examined the **ways in which human beings have organized** themselves to meet their needs – including the urban revolution. Now we will **turn to the first civilizations of the world**. First, those of the **Eastern hemisphere, (the old world)** and then the **Western hemisphere (the new world)**. Today we look at the old world. We'll get an idea of what life was like in those early civilizations and how they compared to one another.

The word civilization comes from the same root as the word city. The **Latin word civis** referred to the citizens, while the form civitas originally meant citizenry – that is, the community of citizens. It meant more than just the plural. It later became the word we know as city.

The development of life within the city versus the nomadic and agricultural communities is the focus we take up here. Cities developed from agricultural communities and we must remember that people within an urban settings are not superior to nomadic people or agricultural people but rather have had the (dis-)advantages of the development and discovery of the city organization to better their lives physically and agriculturally – and with more time for leisure.

**In the 12<sup>th</sup> century, St. Bernard of Chartres said, “We are like dwarves seated on the shoulder of giants. We see more things than the ancients and things more distant. But this is due neither to the sharpness of our own sight nor to the greatness of our own stature, but because we are raised and borne aloft on that giant mass.”** Mario Montessori used to emphasize the fact that earlier people were, in a way, more advanced than later people because they had less to go on to build up their lives. Before there were bridges, there were no bridges! The early people could not look at a bridge and decide to modify the design. They had to invent the entire idea. Later, people improved upon the initial brilliant concept that the water could be spanned and thus safely crossed.

What we want to present the children is this **idea that the mind of human beings continues to work and has taken from what others have discovered.** Human beings have made concrete the ideas they have conceived to help life be lived more easily. However, the minds of human beings have not rested but have gone forward. In the same way, our own minds can do the same thing. We can find out **what has been left to us as heritage, the art, the culture, the inventions and ideas.** We can add to this heritage as we go. Each of us, because we have intellect, has a chance to contribute to others. We can add to the lives of other people through our own service.

Stuart Piggot said, “We should surely not be far from the mark, if we thought of civilized societies as those which worked out a solution to a problem of living in a relatively permanent community at a level of technological and societal development above that of hunting in bands, the family farmstead, the rustic self-sufficient village or the pastoral tribe and the capacity for storing information in the form of written documents or their equivalents. Civilization of all human culture at whatever level is something artificial and man-made, the result of making tools, physical and conceptual of increasing complexity in responding to the enlarging concepts of community life developing in men's minds.”

The **human mind in conjunction with the work of the hand, created civilization.** In his book, “The Origin of Human Kind”, Richard Leaky states, **“We are so very much a product of the culture that shaped us that we often fail to recognize it as an artifact of our own making until we are faced with a very different culture.”** This is the **supra-nature** of which Dr. Montessori spoke. It is the spiritual and physical territories that vary from place to place and from one time to another but upon which all of us are now dependent. We could no longer live a nomadic life of hunting and gathering - unless we were born into such a society. The **vast majority of people in the world are now dependent on civilization.** Keep this in mind as we turn to investigating the life of the people in the four ancient river civilizations of the Eastern hemisphere.

## **Sumer**

**Sumer arose in a land which after two-thousand BC was called Babylon. The Greeks called this area Mesopotamia** – which literally means the land between two rivers. It is the land between the Tigris and Euphrates rivers. And today most of that land is in the country of Iraq.

The civilization of Sumer began around 4500 BC. Agricultural communities grew larger and more productive – creating a food surplus. These communities formed a network exchanging ideas and goods. We talked about agriculture and the possibility of building a surplus in some regions so the labor could be divided. It has been calculated that the wheat yield in 2400 BC, in what is now called Iraq, could be favorably compared with that of the best Canadian wheat growing areas today. Some historians consider perhaps this is one of the conditions which predisposed this region to the development of

an early civilization - probably the first civilized area of the world.

In 1926, an excavation revealed a great cemetery of the Ur of Chaldees (mentioned in the Bible). Royal tombs were discovered within the cemetery which disclosed treasures of gold, lapis and evidence of funeral rituals. Eridu, another city in Sumer, was a ceremonial center for a series of small communities. By 4400 BC, the population of Eridu was thought to be about 5000 people. Archeologists have excavated a mud-brick temple with fairly substantial houses surrounding it. Evidence indicates that the houses closest to the temple were for the elite – probably priests. As houses got further from the temple, it seems the rank of those living there descended, with craftsman living near the temple holding a higher rank, and farmers at the outer ring. From excavation in **Sumer, it became evident that about 3200-3100 BC, there was a complex society with cities, specialized craftsman, cooperative irrigation work, ceremonial centers and writing.** The cities were surrounded by brick walls and a ditch. They were dominated by temples built up on high platforms. Outside the cities were dikes, gardens, walls, canals and harbors. Excavations have led to estimates that at its height, the population living **within the four square miles of Ur reached half a million people.**

The Euphrates river was easily navigable. Therefore, **trade with people along the Persian Gulf and the Mediterranean Sea took place.** The Sumerians also used caravan routes for overland trade. They imported metals, timber, animal skins, ivory and precious stones such as lapis lazuli and turquoise. The Sumerians were master of metallurgy, stone-carving, glass-works, filigree-work and carpentry. Metal working was done in gold, silver, lead, copper, tin and the alloy called bronze.

**Another Sumerian contribution to civilization is the wheel.** It was invented first as a potter's wheel. By 3000 BC, the wheel had been used to make vehicles more mobile. This means the axle must have been invented as well.

The **earliest written documents of Sumer** are domestic or commercial lists for deliveries – such as for bread, beer, cattle, milk, corn and sheep. The **first written documents date from about 3500 BC and give us a glimpse of what was important to the Sumerians to record. It is thought that their writing began as clay tokens used to identify goods for trade. And these evolved into a written script.**

The Sumerians also made important contributions in the area of math. They had a **system of calendars, the base 60 system of our clocks and the degrees for measurements** of fractions come from them. Words still in the English language such as alcohol, gypsum, myrrh and saffron have Sumerian roots. As we are talking to the children about contributions of earlier societies to the development of the life of human beings, at some time be sure to make the children aware of some of the contributions of the Sumerians to our lives. The **Sumerians found themselves in a rich environment and were able to provide themselves with many benefits of a civilized society. The alluvial plain of the Tigris and Euphrates rivers gave them a fertile plain for growing crops and they took advantage of the possibilities offered. They developed their lives around it. It was the genius of the Sumerians that developed the wheel,**

**writing, glass, beer, a calendar and the walled city** – all of which we still have in our lives today, and which come to us from many thousands of years ago.

## Egypt

Egypt of the Nile Valley is perhaps the best known of the four river civilizations. Were the civilizations of Egypt and Sumer able to learn from one another? Did the Egyptian civilization arise on its own from the genius of its people? Or, did they take some things from Sumer in order to begin? This is a question that exercises the minds of historians. They explore the differences and similarities of these two civilizations. We can only look at the evidence and surmise from it.

**The Egyptians, like the Sumerians, created a civilization based on irrigation.** There is a difference in that, in Sumer, the irrigation was done through a system of canals, while in Egypt, the irrigation came about through seasonal flooding of the Nile. The Nile River Valley was also very fertile – and could support a large population.

**Five-thousand six-hundred years ago, the average Egyptian probably lived much as some Upper Nile villagers do today; growing wheat and barley in riverside gardens. They gathered wild vegetables, herded sheep and pigs and hunted and fished. They had clay pottery.** But also some vessels of alabaster and basalt. They were skilled flint workers. It seems they learned the art of the copper-smith from Asia. They made a type of glass called faience. By 5200 years ago, the first trading settlement arose at El-Omari.

Other trading settlements arose and seemed to be the link between the many villages as each village grew and increased their food production. Within just 2000 years, from the establishment of El-Omari, Egypt had become a civilization. There is evidence to suggest the **Egyptians were in contact with the people of Sumer and these contacts helped the Egyptians in the development of their civilization.** Sumerians cylinder-seals have been found in Egypt. Sumerian motifs appear in Egyptian art. Early Egyptian mud-brick buildings seem to have copied the style of building found in Sumer.

**The first Egyptian writing was derived from early Sumerian writing.** Despite all the similarities between the two, Egyptian civilization differed from Sumerian civilization. **When Egypt entered history and became civilized, about 3200 BC, all the country was under one ruler the Pharaoh – who was considered the God incarnate, as King.**

**The Pharaoh ruled over a state six hundred miles long. This was the first state in history.** It was quite different from the separate cities which formed the civilization of Sumer, which only became a nation-state much latter in history. The

**pyramids show a grandeur unequaled at the time and still enduring today.** Glyn Daniels suggests that the origins of Egyptian civilization can be explained in terms of stimulus and diffusion. We can think in terms of it coming from Sumer to an African society in the Nile Valley. It was a society already on the way to civilization when contact was made with Sumer and which in the course of time, might well have gotten to civilization without the benefits of interacting with Sumer. This is one historian's viewpoint; there are other interpretations.

## Indus

A third river civilization of the Eastern hemisphere is that of the Indus valley. This is another civilization which **grew up around a river and flourished between the third and second millennium BC.** In the Indus valley of present-day Pakistan, there were two main cities: one was Mohan-jo-daro and the other was Harappa. The Indus valley river civilization spread over an area much larger than either Sumer or Egypt. It extended **one thousand miles from north to south.** The **two main cities had citadels and ceremonial centers.**

In the center of Harappa lay an artificial mound which was the citadel or stronghold; on this was a large granary. Surpluses of grain then would have been as important as gold is today. Also, on top of the citadel was a large building that seemed to be a combination of housing for the priests and either a ceremonial center or a school. Around the citadel were shops and residences. The streets were laid out in a grid pattern. This rectangular town planning was the first in history. Many of the **houses in the Indus civilization were organized as apartments, equipped with garbage cans and a brick drainage system. (Compared to Europe at the time, this was very advanced.) Houses for the elite had their own baths and toilets.** There were public baths for ceremonial purposes. Where did they get these ideas from which we later borrowed?

The Indus valley civilization was based on agriculture; early crops included barley, wheat, melons and dates. They had also domesticated animals before the rise of the cities. The **most important advancement in agriculture in the Indus Valley was the controlling of flooding through a planned system of dams, dikes and irrigation canals.**

**They had pictographic writing found on seals which have not yet been deciphered.** The seals looked like buttons with a raised ridge pierced by a drill on the back-side. It seems they **threaded a string through the hole and then wore the seal at all times.** On front of the seal was a picture that had been carved into it. At first, these pictures were intricately designed animals such as alligators and tigers. Later, they became more and more stylized. They used the seals to stamp wet clay that sealed the containers of wares and goods. We imagine these were used by merchants and ultimate users of those goods.

The **people of the Indus Valley had extensive trade-relations with the people of Iran, Afghanistan, Mesopotamia and Southern India**. About 2500 BC, pottery, seals and beads reached Mesopotamian cities from the Indus Valley via **Bahrain and the Persian Gulf**. Again, we come upon the question of the origin of the Indus Valley civilization. It was different from Egypt and Sumer, and yet similar in many ways. Was it an offshoot of both Sumer and Egypt? Or, did it arise independently and make the same discoveries and inventions? There is evidence that a few men from the Northwest villages of the Indus Valley journeyed to Sumer – but docked their ships at Dilmun and brought back information from Mesopotamia. This information may have included the idea that villages could be grouped together and then grow into towns and cities – and that an urban life-style could be created. But we don't know for sure. Thus, the pattern of creation of civilization in the Indus Valley begins with the same foundation but is varied, once again, as it develops.

## **Yellow River (Huang Ho)**

The fourth river civilization of the world is the Yellow River civilization or Huang Ho civilization of China. It **began in the lower and middle course of the Yellow River**. The agricultural revolution preceded the rise of civilization in China, just as elsewhere.

The earliest date for the agricultural villages in China is the middle or end of the third millennium BC. These **agricultural communities grew wheat, millet and mulberries. They built terraces along the sides of the steep hills to slow the erosion and to contain water for the dry season. They also built irrigation canals in the lowlands. They raised pigs, cattle, sheep, dogs and chickens. They kept silkworms and had looms for weaving silk. They made clothes from hemp and silk. The Chinese, from the earliest times, were renowned for their ceramics**. We still call fine ceramic dishes “china” today. They were the first people to glaze and paint their pots. They used two types of clay. The coarse everyday pottery was made from gray clay. But they applied to it glazes containing metallic elements which when fired, became iridescent. The other clay was white; its very fine texture resulted in thin translucent ceramics called **porcelain**. One craft particular to the Chinese people was **bronze metallurgy** – which began with the making of ornate ritual vessels. They used the **lost-wax method of making molds**.

The Chinese people were very interested in predicting the future. They would ask their Gods, “Will it rain? Will the crop fail”? The seers scratched or painted the answer onto bones or tortoise shells. These oracle bones were the precursor to Chinese writing which continues to be characters today, represented as one character per word or idea. **Chinese writing has been used in its original form longer than any other written form**. Once again, we see similarities among scripts.

The Yellow River civilization grew from an agricultural community into cities. The style of working bronze is particular to China. Their beginnings and the **uniqueness of their bronze work has led to the theory that the Yellow River civilization developed independently**, not borrowing from other early civilizations.

## **Current Explorations**

We may want to look into “Looking for Dilmun”, by Geoffrey Bibby. There is some evidence to suggest that Dilmun predates Sumer as a civilization. Dilmun started on the island of Bahrain in the Persian Gulf. Archeologists had begun to excavate Dilmun on the coast of present day Iran. But the digging would have taken them beneath the woman's bath-house and the sheik would not give permission to continue excavation. The excavation thus-far has suggested that there may have been contact between Dilmun and all the early civilizations.

## **Epilogue**

Tell the children stories about these ancient civilizations. Build up a picture of these people who lived there. Throw out questions to the children that cause them to speculate about the origins of each civilization. Guide them to look into how these civilizations were similar and different. This is what the gift of their reasoning minds provide.

“Guns, Germs and Steel” is another book by Jared Diamond, that is useful in researching this subject-area. He provides much information and has his own ideas of what happened during development of these civilizations.

Grow a papyrus plant in the classroom if possible. Relate this to early writing. Children with limited funds can rely on parents and outside sources for things that follow the theme of – Egypt, for example.

January 18, 2005

## **Early Civilizations of the Americas**

As we explore the early civilizations of the Americas, notice similarities or major differences from the early civilizations of the Eastern hemisphere. Did the various accomplishments of each civilization arise independently? Or, were they borrowed from others? This is what we want the children to notice as they explore civilizations.

### **As the Spanish came to the Americas, they met 3 ancient civilizations.**

As in the old world, and with human beings from the beginning, the earliest people were hunters and gatherers. At some point, food production was begun. Effective food production in the Americas came about over a period of many years – between 6500 BC and 1000 BC. The basic crop of village farmers was maize (corn), which was unknown to the Eastern hemisphere until after European contact.

**The three main early civilizations of the Americas should be presented** to the children. The stories we present may need a bit of judicious editing. All of the children will undoubtedly find information and be fascinated by some of the more gruesome aspects of the rituals of the Aztecs and Mayans.

Here are a few points about these civilizations to get us started. This is one area of history where new discoveries are constantly in the news – so be sure to keep up on the newest information available and make the children aware of these sources as well.

### **Aztecs**

The Aztec civilization was founded and flourished in what is now called Mexico, beginning in about 1425 AD – and yet was a well-developed and dominant civilization when the Spanish arrived in the interior in 1519. We can begin by showing the children a map of the region around Mexico City as their originating territory.

Despite being a settled people, the Aztecs were warlike. They conquered, sacrificed and enslaved them for expansion of lands and control of goods. They were skilled artists, working with gold and copper – as well as stone. The Aztecs built pyramids. The pyramids were not tombs like those of the Egyptians, but were temples used for ceremony and ritual. In this way, the Aztec pyramids were more like the Sumerian pyramids than the Egyptian's.

They used a type of hieroglyphic writing. It was a combination of ideographs and phonetic representations. Encourage the children to take up a comparison of the Egyptian and Aztec hieroglyphs. Was there contact and sharing of ideas? Or, was this independent invention?

The Aztec number system was a base 20 system. They used place value and added zero. They would attach the number symbol directly to an illustration of the goods. For example, a small flag represented a 20 unit. So if you had a honey jar with 3 flags drawn on it, it meant 60 units of honey. Show them a representation of how this might be done – using rectangular flags. Encourage the children to compare other number systems to this one. When did other civilizations use place-value or add the zero?

The Aztec civilization built up slowly from a hunting and gathering economy. They invented a farming system called **chimampas – meaning floating islands** – which utilized the marsh lands. With successful crop growth, they began to settle and establish communities. These farming communities gave rise to urban communities which grew into the highly organized government of the Aztecs.

We see similarities with Egypt and Sumer but there are differences too. There is increasing evidence to show that behind the rise of the Aztec is an **earlier Middle American civilization called Olmecs. This civilization of Mexico, situated west of the area of Maya, flourished from 800 BC to 400 BC**. This area differed from the alluvial plains of the old-world civilizations. The area of the Olmec had very heavy annual rainfall. It was situated in a high-lying tropical forest with a great deal of swampy land around it. The Olmec were great stone carvers making massive figures. They also carved small jade objects. The Olmec writing was in hieroglyphic form which has not yet been decoded – just as the Indus Valley writing has not yet been decoded.

## Maya

The beginning of the Mayan civilization has been traced back to the 5<sup>th</sup> century BC. This civilization of Central America lasted until the 9<sup>th</sup> and 10<sup>th</sup> centuries AD when for reasons yet unexplained, the Mayans abandoned their cities, leaving incomplete

buildings and other unfinished work.

The Mayans covered a vast territory, from south, from what is today Mexico and stretching to what is today Guatemala and Honduras. The Maya were fine builders. The architectural structures incorporated figures right into the buildings, statues and other stone works, as well as frescos (painting on plaster incorporated right on the wall) that survive today.

There was no metal work except for a few decorative pieces of copper and gold. The fine workmanship of their architecture and art was achieved using only stone tools. Once again, the Mayan culture was an agricultural society. Their **farming system was based on 3 crops: beans, squash and corn. They planted these using a hoe as garden plants.**

Remember that the old world had serial crops cultivated by the plow. The new world had garden plants cultivated by the hoe. How does the environment impact on this difference? Was this purely a matter of inventions? Get the children thinking and exploring around these differences.

## **Incas**

**When the Spaniards invaded Peru in 1553, they found the vast empire** of the Incas. This empire extended from what is now Ecuador in the north to about the middle of what is today Chili in the south – **more than 2500 miles in length.** This vast territory was all under the rule of one government – the Incas.

The **Incas had no form of money or writing.** However, they kept accurate accounts by means of knotted and covered strings which served as memory aids. We might want to rethink the idea that writing is usually considered one of the cornerstones of any civilization. Their agriculture was based on extensive terraced gardens built into the sides of hills and mountains.

Their metal work was extensive and included techniques such as alloying, casting, soldering and gilding. They manufactured weapons and armor of copper. They used highly advanced techniques in weaving and other textile arts. They built step pyramids with flat tops. They also constructed massive palace complexes and had large building complexes with corridors and courtyards but no true apartments – even though these complexes were in areas of large populations. These are thought to have been government office buildings.

## Epilogue

The question to engage the children's minds is this: where do the new world civilizations come from? We discussed contacts and exchange of ideas in the old world civilizations. We noted that some scholars think there was contact between civilizations in the old world, while others do not. What about the new world? We need to phrase our stories in the realm of speculation: "it would appear that", "current evidence shows that".

Written records do not exist for the time period of the rise of the new world civilizations. So it has to be surmised and not considered fact. Archeological research in the Americas seems to point to the independent rise of civilizations there. The patterns differ in many ways from the old world – and in other ways they're the same.

Tools and weapons are mainly made of wood or stone. Occasionally, tools of copper and bronze were made before European contact – though no iron is found. Another difference is that the people in the civilizations of the Americas did not use the wheel – neither in vehicles nor in pottery. Strangely, they had a few small toys with wheels. They had agriculture but no cows or breed-stock. They had domesticated small animals like chickens. They used sticks and hoes for digging and had no plows.

Some civilizations had a form of writing. The Aztecs had a form of picture writing that was never further developed. The Maya used their form of writing only for religious and astronomical context. The Incas had only the knotted strings.

What about religious differences and similarities. All the religions of the Western hemisphere presented a belief in life after death. The Mayas believed that there was a long dangerous journey through an underworld before reaching heaven. Egyptians believed there was a similar journey.

The significance of the presentation to the children of the civilizations of the Eastern and Western hemispheres lies in the fact that they were a huge step forward in the life of human beings which came about through cooperative living. Tasks and labor could be shared; society was organized in such a way that the sharing of tasks benefitted the individuals.

As the children study the various early civilizations, they can see the patterns and begin to contemplate: was it agriculture that enabled society to organize and "grow up" into urban life? How was the division of labor necessary to this development? A.L. Koeber, an American anthropologist (1876-1960) said, "We must consider that civilization is an inevitable response to laws governing the growth of culture and controlling the man-culture relationship."

We should remember that human beings have always had to satisfy, first, their physical needs. Once people satisfied their physical needs, they began to satisfy their spiritual needs. Once they began to satisfy their spiritual needs, the first experiments in civilization came about. This leads to the first attempts at literate urban society. We ask the children and ourselves to study these ancient civilizations of both the Western and Eastern hemispheres, even though they are so far removed from our own times, to **help the children contemplate the process of the intellect of human beings, to assist them in thinking about the process which led to creating what was not there – achieving a fuller life and creating a fuller society.**

Remember the History Question charts form the foundation for the study of the life of the people for your presentations and the children's exploration of any civilization and any group of people, whether food gatherers, agriculturists or civilizations. We should keep abreast of any new discoveries in archeology and bring those to the attention of the children. The easiest way to do this is to subscribe to magazines, such as The National Geographic or the Smithsonian. The Andes and other mountain tops experience melting and bodies are often found, still clothed and so on.

The National Graphic from October 1988 (vol. 4) has an article about the Moche; this article provides an idea of how artistic and advanced the Moche people were. The Moche 100 BC to end of 100 AD, agricultural society that flourished in the desert regions between the Pacific and the Andes. They raised huge monuments of adobe bricks used as tombs for their royalty. In this way, they were similar to the Egyptians. Also like the Egyptians, they placed gold and fine pottery in the tombs for the afterlife of the royalty. One theory about one particular tomb that was found un-looted is that the person buried there was a ruler, warrior and priest – all in one.

The Moche artwork discovered in the tomb points to art and technology that rivals the Maya, who were contemporaries of the Moche. It appears the Moche had pottery and weaving techniques about 1200 years ahead of the Incas. The Moche did not develop an empire like the Incas but they did rule over a 220 mile stretch of the coast. Mostly, this consisted of small settlements in the mountain valleys like those of the Sumerians. Also like the Sumerians, the Moche irrigated their fields using channels and canals. They supported about 50,000 people in this manner – greater than the number of people living in that region today. So did the Moche have contact with the Maya or others – or was the rise of their civilization independent? It is thought they traded with people as far away as present-day Ecuador. **The Moche did not have a written language nor were their lives written up by foreign visitors as were lives of the other groups.** At the present, all we know about them has been discovered from artwork which shows, in detail, ritual warfare. No doubt there will be new findings to come.

Remember, the **stories or presentations are based on the History Question charts**. These should lead the children to doing their own explorations for more information. Then they can record their own work in writing, drawing, modeling, timelines or whatever other form they like. Not every child will do this for every civilization presented; but each child should become intrigued at some point and thus inspired to find out more. We need to keep giving stories!

January 18, 2005

## **Time**

Another area to be taken care of in the elementary class in relation to history is to help the children understand the time scales used. So far, the work we have done with dates is not very specific. On the timeline of life, we talk very little about dates. The two timelines of human beings have some time indication but they are very mostly general. At a certain point, the children should be able to put history into the perspective of recorded dates. This means there are aspects of time we must help the children understand.

### ***Clock***

The children need to be able to read an analog clock (circle face with hands). We have clocks available from Nienhuis. We can begin by reviewing the work with the sun and earth. How long does it take the earth to rotate?

#### **Begin**

Here I have a clock. Right up here is where we start; it is called 12 o'clock. Then we have 1 o'clock, 2 o'clock (continue, each hour, place a red number on the points on the face) ... and back to 12 o'clock. That is the first half of our day!

(at some point, count around the clock using the minute hand as well, counting by ones and then skip-counting by 5s – emphasizing 60 minutes in an hour as well as how we typically read minutes on the clock)

This long hand of the clock, we call that the minute-hand. The short hand of the clock is called the hour-hand. A day is divided into 24 hours. It takes 1 hour for the minute-hand to go all the way around the circle of the face. In that time, the hour-hand moves from one number to the very next number. (we could work in background information from the Babylonians).

(move hands to accord with dialogue)

That's 1 hour, that's 2 hours – this says 2 o'clock. Another hour and now the clock says 3 o'clock. (continue around the entire face of the clock) We write this "o'clock." It's a contraction meaning "of the clock."

The minute-hand goes around the clock, minute by minute. There are 60 minutes all the way around. So that's 1, 2, 3, 4 and 5 minutes between each number. See how little our hour-hand has moved during these 5 minutes? I want to know what time it is (12:15)

We can count by 5s, right? This says 5, 10, 15 minutes after 12 o'clock. (If written formats are already presented say, "We write that 12:15")

(do second period)

I would like you to make three-fifteen on the clock. How about four-forty-five?

(get to the point where you set the dial and they tell what time it is)

(move to 10:30)

Let's look at this. The minute-hand is half way around the face of the clock. Let's consider our fraction of  $\frac{1}{2}$ . We can say this is half past ten. (hold up the  $\frac{1}{2}$  circle fraction piece)

(repeat for quarter past and quarter until using the fraction insets to help solidify)

## Epilogue

Let them practice this for a substantial period. Show them the format "6:00" - saying "this 6 is for the hours and the zeros are for the minutes." If you have a real clock with minute markings that is useful. Some children may have studied the analog clock in the primary classroom. At some point, help them with the particular terms.

"My Day" (using a clock stamp and a wider adding machine paper roll), is a material source of practice. While unrolling the paper, have a conversation with the child. "What time do you get up"? "What time do you go to lunch"? Then write "My Day - Eric" and take the stamp pad, stamp the times (in order) along with a phrase that describes what occurs at that time of day, onto the paper roll. Make a little decoration between each event - time progressing from right (early) to left (late).

One of the things they have trouble with is if you say 4:45 they think the hour-hand should be on the 4. Help them to see the minutes influence the position of the hour hand - in between numbers on the face.

If you teach in a culture that uses a 24-hour clock, show them blue tiles that represent those hours instead of the red pieces we used to populate the clock face; go around once using the red numbers and then use the blue numbers on second 12-hour round.

Children will need the clock for Clock of Eras and in others circumstances.

## ***Calendar***

This may have begun on the primary classroom. We may have recorded planned events on the calendar already as well. Help them learn the names of the months and their order, names of the days and their order. We can tell stories about their names.

### **Begin**

January comes from the Roman God Janus. Janus had two faces so he could look forward and backward at the same time. And so he begins the year.

October got its name from the word for 8 – the number 8. But it's our tenth month! How did this happen?

Thursday comes from the word Thor – which is Old English.

### **Epilogue**

Have different kinds of calendars – one system only - one with the whole year, one week by week, month by month, day by day. Show them various ways to write the dates: using words, using numbers, using different number ordering such as that used in Europe – though this is after they have mastered their own culture's most common usage. Finally, older children enjoy calendars from other cultures; e.g., Jewish, Chinese.

## ***Timelines - My Life***

When we are dealing with time, we deal with a scale. Each of the calendar styles has scales – a year, a month and so on. A timeline is another scale for time; to help the children understand this, have them make a timeline of their own life. First, help them decide on what scale to use – so that each year of their life will have enough room for a picture of some aspect of their life. Begin with an estimation of large the picture will be.

### **Begin**

Will we need 6 inches for each picture? How old are you? Your six – so how many inches will we need? (let them calculate) And then we must add a bit more space for our title. So how long, altogether, will your timeline be?

(cut off that piece from the roll of paper – help them to mark space for each picture)

Let's put the date at the bottom for each picture – from when you were born until now. You can ask your parents for stories and even photographs about something for each year of your life. The you can write something short about this event and put it on the timeline – like “I went to Texas when I was 3.” Maybe you want to do one for your parents of grandparents after you finish your own!

## ***Timelines - BC and AD***

We need to help the children understand BC and AD. Take a narrow roll of adding machine paper – about 2-3 inches wide – and decide on a scale for the timeline. The timeline should goes from about 2100 BC to 2100 AD. Ask them how long this timeline is? And how far should we make each mark in years? May 100 years – and perhaps one inch for each hundred years; how long would it be? What about if we made each mark 6 inches (allow them to calculate and plan the timeline dimensions).

Start with 2100 BC and make a mark and write a corresponding year. Work from right to the left, making marks along the way. We mark zero even though there is no year zero. Don't laminate it – let the children remake it as needed – and keep one on the shelf.

Once the timeline is finished, it can be laid on the floor and the children can place at the correct location on the timeline, pictures that they have collected or drawn of the history of the fundamental needs to human beings and various related inventions.

Children can also make their own scrolls in chronological sequence for inventions or virtually anything else – hats through the ages, for example. Try to get them to make their own illustrations on these timelines, hand-drawn for hand-eye practice.

## ***Timelines - BC and AD***

In the 9-12 class give the children Hebrew, Russian, Chinese and other calendars – these are still in use. Give them the Mayan calendar for an example of an ancient calendar. Introduce Common-Era (CE) as it comes up – as well as Before Common-Era (BCE) and After Common-Era (ACE). Introduce the same information for Latin BC (before Christ) and Anno-Domini (meaning “year of our Lord”).

January 19, 2005

## History Timelines

History timeline work is, in a way, a culmination of the children's history work. We've been following a sequence that rests on the development of human societies. The development of those societies rests on a fairly simple formula: how does this society, as a community, make for themselves a physical and spiritual territory?

In some cases, these studies can go chronologically, as we look at them from their beginning until their decline or disappearance. In some cases, these studies will go horizontally, comparing one to another in various regards. In other cases, we can mix the two kinds of studies – vertical and horizontal.

There is also a regional dimension to any society one may look at. Were there any other societies like this one – in other regions – at the same or at different times. What did they make use of from their environment to make their lives satisfactory – both physically and spiritually?

If we keep these ideas in mind, this theme we've talked about all year, then it does not so much matter which societies we present to the children for them to investigate. As long as we're following a sensible logical order, the work will have meaning for the children. It cannot be presented haphazardly because this will confuse them. So we need an approach that is rational. For example, we might do several river civilizations before going on to a new society.

It also does not matter where the children will look for more information after our presentations, nor what they explore nor if they explore – as long as we keep the public school curriculum. That curriculum is our only gold-standard.

There are two quotations that form the motif for this exploration of societies. The first is from the “Ascent of Man”, page 437, by Jacob Bronowski, where he says, “We are nature's unique experiment – to make the rational intelligence prove itself sounder than the reflex. Knowledge is our destiny.”

The second quote from “The Universe is a Green Dragon”, by Brian Swimme, page 19, where he writes, “The human species is the youngest freshest most immature newest species of all the advanced life forms on the planet. We have only just arrived. If we can remain resilient, if we can continue our questioning, our developing, our hoping, if we can live in awe and in the depths of wonder, we will continue moving into the only process that now matters – our authentic maturation as a species. It is in this way and only this way that we will enable the earth to bloom once again.”

These ideas are what we've talked about all year, as we've talked about history. We have to keep on reminding the children of what we are as human beings – what human beings before us have done and then what each of us can do for our part. This is why history holds such a central place in cosmic education. It has to be at the

center of our curriculum and the work we're doing in our classrooms.

**We have certain facts to keep in mind.** If we are to prevent history from becoming mere dates, events and names to be memorized – learned because they will be on tests later on and otherwise divorced from any meaning to our lives today – we have to, first of all, give the children an idea of all of the preparation that had to take place before the coming of human beings as well as why all that was important. What that preparation entailed and what that environment holds within itself are germane to that question. With the advent of human beings, we attempt to present to the children some idea of what earlier lives were like – when people first came on the earth. Through those sequential phases of history, how has human life developed – to arrive at what it is today.

Along with this, the impact of human beings upon the environment needs to be looked at. The changes that we have made to the environment and our dependence on it – and our inter-dependence on environment and each other are instructive

As we move through recorded history and come to present times, **trade and commercial relation charts** are used to give indications of what the life in one's own country is like today. These trade relation charts are to do with the needs of human beings and how those needs are being satisfied today. Economic geography ties up and pulls together all of the geographical factors that we will study – as well as human relations and human life. Trade and commerce don't just relate to the here and now.

Trading between groups of people has been going on virtually since the first different groups came into contact with each other. So they've been a part of human life for a long long time. We will also talk about what that early trading and commerce was like as we tell our history stories. We use the **History Question charts** as the foundation for our presentations in history – both in our presentations to the children and for the children's follow-up work. These charts also include questions about trade – so trade is one of the universal factors of history we will explore.

This process of using the History Question charts focuses our thinking around the needs of human beings and how various groups of people in the past have found different means to meet those needs. In dealing with the needs of human beings, as the basis for the growth of societies, we also need to make sure the children are aware of idealism – the core of the spiritual territory – and that that idea or spiritual territory is more important than the physical territory.

Throughout time we can find many groups of people who left their physical territory behind in pursuit of spiritual territory – or took their spiritual territory with them.

When we come to history timelines, which begin to show the stories of societies in a chronological sequence, **there are two types of timelines. One type is to give an impression of a society** – how they lived, worked and played, during a particular time-span. These timelines are not the kind used for memorization. This kind of timeline is developed by the children after a presentation or a series of presentations given by the teacher. They can also be developed by the teacher and children working together. One of the most effective ways to introduce this kind of work is to work with them as part of that group – as a peer. We teach them how to do this work by doing it with them.

The **second kind of timeline used for memorization** are the kind used to help them learn the required aspects of the public curriculum. With both types of timelines, the guide for the presentation of the lesson(s) is the History Question charts. The questions used there and their answers ought to make it possible for us to give the children a clear vision of what the society was like; what the people did and what their environment was like.

The History Question charts also guide the children in their further explorations and work. With the Fundamental Needs charts as a basis and the History Questions charts as the structure for the lessons – a picture can be built up for the child of the successive societies that have developed. We can begin with the stone age cultures, the hunters and food-gatherers – to the pastoral people – the agriculturalists – and finally to the urban dwellers.

Along with that view of the progression of societies should also come the possibility of bringing the child's attention a horizontal look at history – within a certain time-span – looking at several different society's development. History is not neat and tidy and shouldn't be if it reflects reality.

For all with a traditional educational background, especially if one has studied history in the chronological approach, it can be difficult to let go of that – to initiate these comparative studies we are not experienced with. Especially in the beginning, we should remain vigilant to stay true to the Montessori method. If we let go of the children, they will lead us; so **don't force our backgrounds, biases and partiality on them.**

We are there to help the expansion of the child's intellects. We are to aid that free exploration. We have to help them think – in the context of history, that means the story of the beginning of human beings coming into the universe – from the beginning through and up to urban life and the industrial society. It is leaving the child free, after we've given the basic presentation(s), that enables them to understand that the ways of life of different societies have come about according to the basic needs of all human beings. It is the building up of their own physical and spiritual territories which differ among individuals and societies – because they met those needs in a different way. It is in this way, we help the child develop respect for other people, traditions and cultures.

**We will never be able to get people to live together peacefully by lecturing.** Appreciation has to come from understanding – putting oneself in the shoes of another. This is becoming more and more important as societies have become more

diverse and increasingly at odds. We need to be able to live together peacefully and respectfully – and sincerely. We therefore need to understand each other. It's the children who can reach this point of true acceptance much more easily than adults can. If history will have meaning to the child, it must be put into a social context. This is what our stories and presentations must do.

**With respect to American history**, or for any other country a teacher may return to, in whatever country we are in we have to something about our own country's history. This is generally a public school requirement for elementary children. In the early 1970s, an American history timeline was developed by a fellow named Sanford Jones; he had worked with Mario Montessori on this timeline. This timeline was tried out on an experimental basis in elementary classrooms around the country. From experience, it appeared revision needed to be made but were never done. We will see this timeline around perhaps in some classrooms. It was not one continuous piece but rather a series of panels that could be placed end-to-end. It extends from pre-Colombian history up until the present. This is one possibility for an American history timeline. The Elementary Alumni Association has done quite a lot of work for ideas for an American history timeline.

One of the difficulties is deciding how to break down our country's history. If we study England, it broken down by ruler typically. But our history cannot be broken down in the same manner. Some of the presentations done by EAA have taken an economic approach, while others have taken an approach according to conflicts. Or they could be broken down on social terms. Our country's history is relatively new and therefore we don't have a perspective of time that tells us what are the most critical things to recount.

A Montessori teacher from St. Paul named Larry Schaeffer, has developed an American history timeline. He was a history professor before that during the 1960s. He is principal of the twin cities Montessori school founded by the Gates Foundation. There are pros and cons to this timeline also. They are for sale but they are not cheap.

See books “People's History of the United States” by Howard Green and “Lies My Teacher Told Me” by James Loewen, for excellent resources. We can illustrate that history is always written from a specific person's perspective.

January 19, 2005

## Timelines for Memorization

Just as we said in other subject areas, we always have to balance the children freely exploring, whatever they are interested in, with meeting the demands of the public curriculum; we want them to be prepared when they leave us – knowing what society expects them to know.

It was in India during the second world war, when Dr. Montessori developed many of the materials and ideas regarding elementary children. In 1944, Mario Montessori said, “Regarding the memorization of historical events, a sort of game has been devised. First of all, the teacher gives the lesson. Then, by way of testing the memory, cards and charts are prepared.”

The teacher giving the lessons is what prepares the mind of the child for whatever facets of history are being investigated. The **information given on those presentations is built up around the History Question charts and the Fundamental Needs charts**. Depending on the topic it may be one or a series of presentations. The lessons given need to include all of the factors that the children are required to know. Dealing with public requirements we cannot leave it up to chance to encounter the required information. We must include all required information in some manner. Hearing it one time is not enough either; **there must be repetition**. So now we must have material that will allow the child to work with the information again and again.

The system that is used in many ways resembles the material used to help the child learn math facts. There is, first if all, all the **material from which they initially get the information**. Once the information has been given, then there are charts the child can use to solidify that information – eventually using it to test himself.

In our preparation for our lessons, we would choose questions from the History Question charts that enabled us to include information which would gradually build up for the child a picture of the social life, geographical region, the spiritual and physical territories built up and so on. After our presentation(s), we would still encourage the child to look further – to find out more.

From the teacher's lesson(s), the **teacher and children then work together** to develop a timeline that includes pictures (preferably drawn) that are pasted right on the timeline. Get in the habit of drawing picture separately for efficiency. Relevant dates should also be included on the timeline. Names of relevant people of history as well as appropriate maps are included per the public curriculum. There would be a colored strip across the top with the name of the group or society charted on the timeline. Beginning and ending dates are included on the timeline.

This **first timeline** is the full timeline – a complete timeline – complete, though the 50,000 foot view. There are 3 additional timelines now. The second timeline,

made to the same scale as the first, still has the colored strip across the top with name of group, beginning and ending dates and intervening dates as appropriate, **the dates of events as appropriate to the public curriculum. What is not on this second timeline** are the pictures or printed information that was on the first timeline and that match the dates. This information would be on a separate set of cards which the child would match up with the dates on the timeline. They could check their work against the first timeline for control of error.

The **third timeline** is made to the same scale with a colored strip, name of the people involved, beginning and ending dates again. But **on this timeline are all of the pictures or stories of events**. The card information from the second timeline is now on the third timeline. The **dates for the pictures are now not on the third timeline**.

The **fourth timeline** is made to the same scale with the top colored strip and name of people involved, beginning and ending dates – but this is the **empty timeline**. Along with this timeline, all of the dates and all of the pictures and stories would be on loose cards. Now the children would build up the whole timeline. Their control of error is again the first timeline.

Timelines like these can be constructed for whatever needs to be memorized in history. It would be very appropriate if you are teaching in the United States, to develop these timelines around one's state history or the history of the United States or for whatever other type of history required.

## Notes on the Construction of Timelines

As mentioned before, the teacher has to get the child started making timelines. They are not likely to initiate this themselves if they don't understand the idea of a timeline yet. This is a possible follow-up to presentations in which the information does not have to be memorized – where one can simply follow his interest.

These sorts of timelines are made singly – not set or collection required. **Like the full timeline (first) it will contain everything.** The first thing to consider is what prerequisite knowledge the child will need to undertake this work; the teacher must take care of that work before beginning to make timelines.

First, the need an **idea of scale**. What physical length on the timeline will represent 10 years, for example. They need to be prepared with respect to **note-taking**. We must have given them presentations on note-taking. They need to have had **previous work with the History Question charts** as well as presentations by the teacher that begin to build up a picture for them, of the society being investigated. **These are not going to be timelines undertaken by a 6 year old new to the elementary class.** They must have learned about pre-history, lessons on present-day societies and on needs of human beings. This might be undertaken by a whole group, to which a lesson has been given – or – it may be undertaken by certain members of that group.

Sometimes the whole group is enthused at the end of a presentation but get started only to find they want to pull out of a group. Let them out for good reason; **let the child try to honor a commitment whenever possible.** Another thing that can come up with projects is that they drag on too long; at times, the teacher must help it wrap up. Meet with them, request status and give perspective about completion. Sometimes, when the group is getting started, we need to keep an eye on what sort of chunk (time-span, period of history, complexity of circumstance and so on) they are undertaking. **Projects have to be scaled for success.**

You may need to remind the children that **the time they spend needs to be recorded in their work journals** as part of the project activity. The only thing that can be traced – if necessary – is a map. Any other illustrations are done free-hand – no photocopier. Drawings should be drawn freehand separately and then glued on.

The specific steps to get started to help them get started:

- Determine the beginning and ending dates
- Determine region and people to be investigated
- Decide on a scale
- Measure the needed length of paper and cut it off the roll

- Color the strip at the top according to scale
- Put beginning and ending dates on the strip
- Date other events of interest
- Put maps that indicate the extent of the territory as appropriate
- Place name of society in the top strip
- Study History Question charts: as a group decide what question are the focus
- Decide who will look for answers to a given question – organize labor
- Do the research per plan and find the answers
- Write the answers in beautiful handwriting and illustrate as appropriate
- Glue answers on the timeline in appropriate locations
- When complete, present the timeline to others